LEADING EDGE INTERNATIONAL AVIATION ACADEMY INC.

PRINCIPLES OF FLIGHT

INTRODUCTION

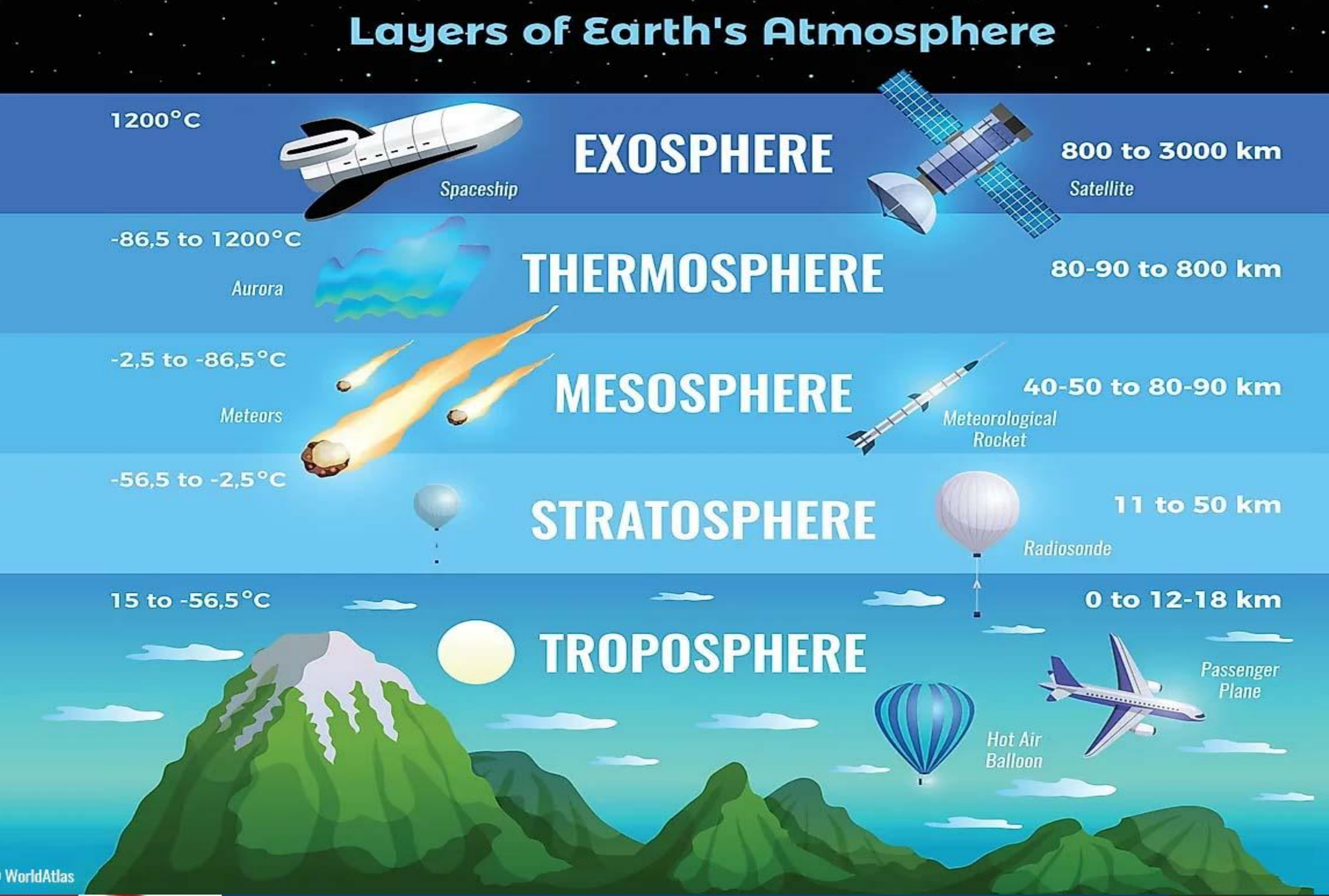
This chapter examines the fundamental physical laws governing the forces acting on an aircraft in flight, and what effect these natural laws and forces have on the performance characteristics of aircraft. To control an aircraft, be it an airplane, helicopter, glider, or balloon, the pilot must understand the principles involved and learn to use or counteract these natural forces.

OBJECTIVE

- The objective of this lesson is for the students to learn about short history of aviation, airflow, and most importantly the aerodynamics

OVERVIEW

- ATMOSPHERE
- AIRFLOW AROUND A BODY
- SUBSONIC
- AIRFLOW ABOUT TWO DIMENSIONAL AIRFOIL AND THREE DIMENSIONAL FLOW AIRFOIL
- FOUR FORCES OF FLIGHT
- FLYING CONTROLS
- TRIMMING CONTROLS
- FLAPS AND SLATS
- STABILITY
- LOAD FACTOR & MANEUVERS
- THE STALL
- AVOIDANCE OF SPIN
- STRESS LOADS ON THE GROUND


Layers of Earth's Atmosphere
1200°C
Spaceship
EXOSPHERE
800 to 3000 km
Satellite
-86,5 to 1200°C
Aurora
THERMOSPHERE
80-90 to 800 km
-2,5 to -86,5°C
Meteors
MESOSPHERE
40-50 to 80-90 km
Meteorological Rocket
-56,5 to -2,5°C
STRATOSPHERE
11 to 50 km
Radiosonde
15 to -56,5°C
TROPOSPHERE
0 to 12-18 km
Passenger Plane
Hot Air Balloon
WorldAtlas

STRUCTURE OF ATMOSPHERE

- The atmosphere is an envelope of air that surrounds the Earth and rests upon its surface. It is as much a part of the Earth as the seas or the land, but air differs from land and water as it is a mixture of gases. It has mass, weight, and indefinite shape

- The atmosphere is composed of **78 percent nitrogen, 21 percent oxygen, and 1 percent other gases**, such as argon or helium. Some of these elements are heavier than others. The heavier elements, such as oxygen, settle to the surface of the Earth, while the lighter elements are lifted up to the region of higher altitude. Most of the atmosphere's oxygen is contained below 35,000 feet altitude.

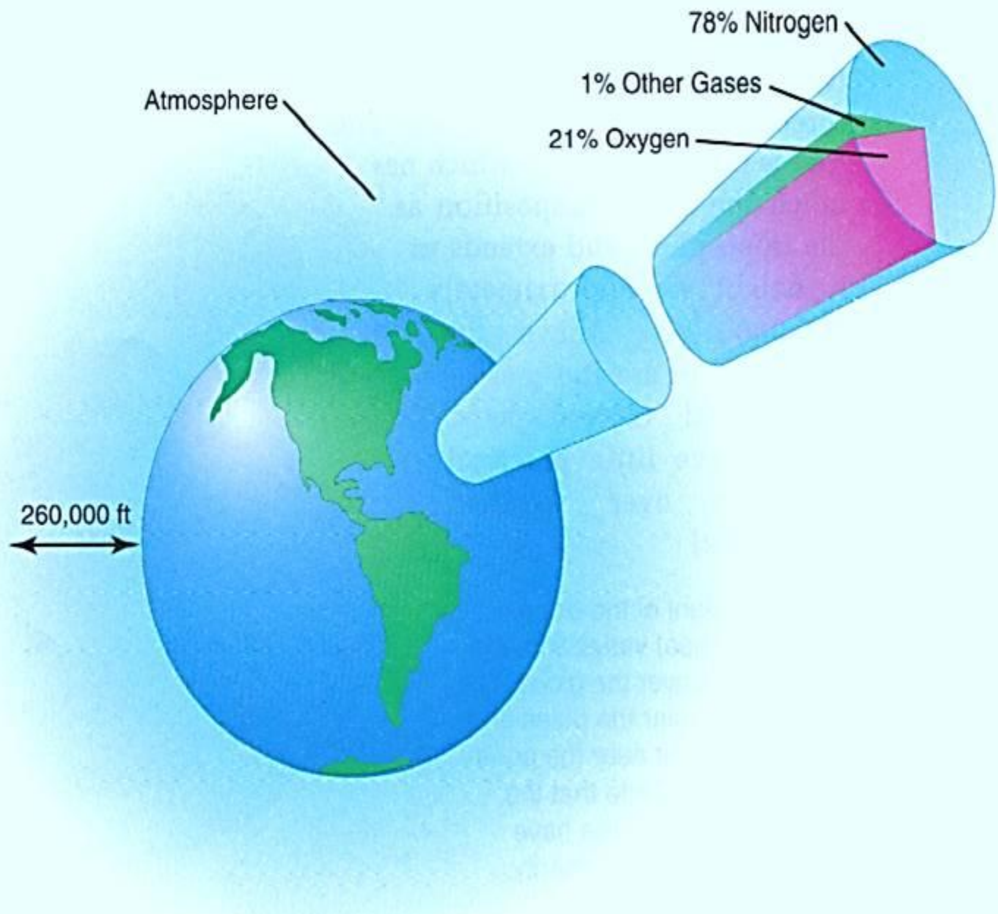


Figure 6-3. The atmosphere is a mixture of gases that exists in fairly uniform proportions up to approximately 260,000 feet above the earth.

Atmospheric Pressure

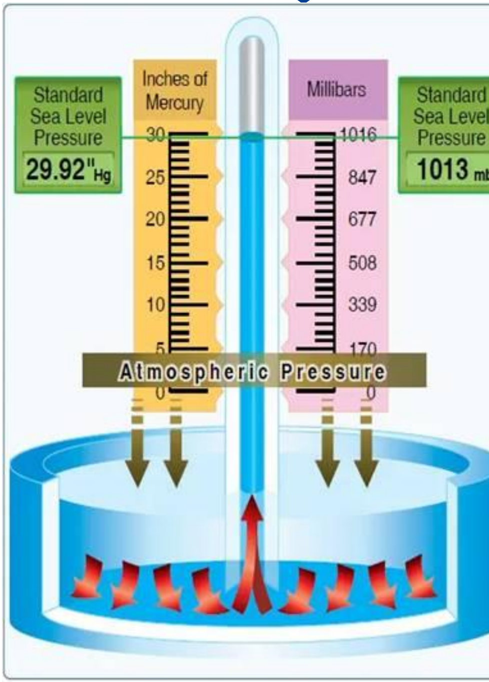
Air is very light, but it has mass and is affected by the attraction of gravity.

Therefore, like any other substance, it has weight, and because of its weight, it has force.

Since **Air is a fluid** substance, this force is exerted equally in all directions. Its effect on bodies within the air is called pressure.

The standard atmosphere at sea level is a surface temperature of 59 °F or 15 °C and a surface pressure of 29.92 inches of mercury ("Hg) or 1,013.2 mb.

A standard pressure lapse rate is when pressure decreases at a rate of approximately 1 "Hg per 1,000 feet of altitude gain to 10,000 feet. ~ *ICAO*



AIRFLOW AROUND A BODY

Air is a Fluid

When most people hear the word “fluid,” they usually think of liquid. However, gasses, like air, are also fluids.

Fluids take on the shape of their containers. Fluids generally do not resist deformation when even the smallest stress is applied, or they resist it only slightly. We call this slight resistance viscosity.

Fluids also have the ability to flow. Just as a liquid flows and fills a container, air will expand to fill the available volume of its container. Both liquids and gasses display these unique fluid properties, even though they differ greatly in density.

Viscosity

Is the property of a fluid that causes it to resist flowing. The way individual molecules of the fluid tend to adhere, or stick.

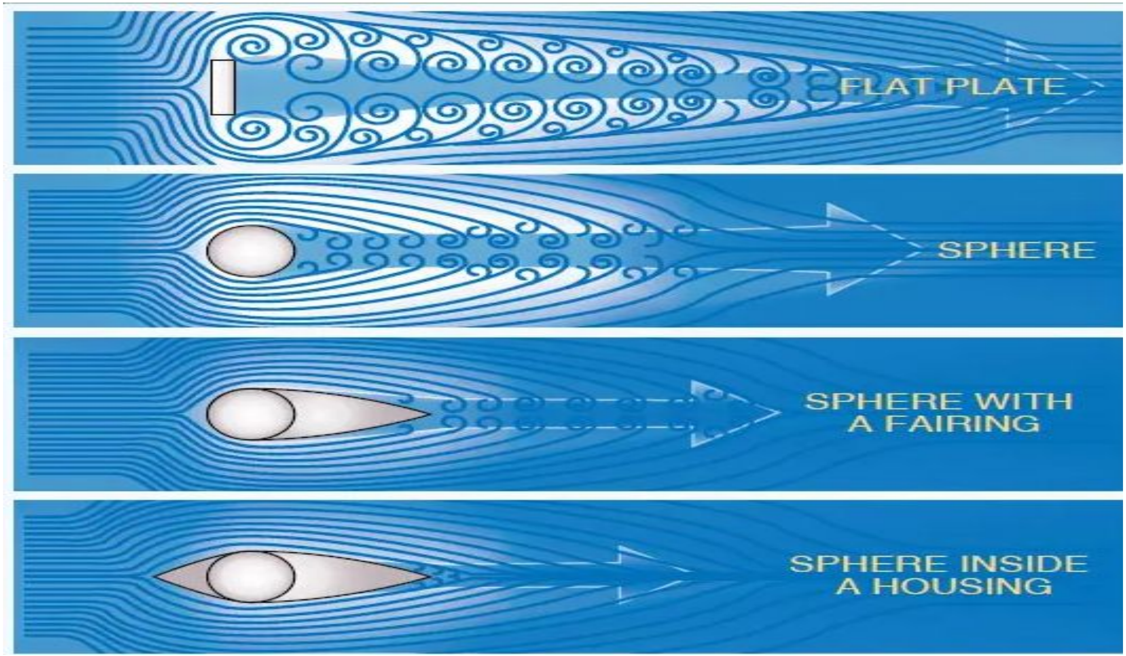
Friction

Is the resistance that one surface or object encounters when moving over another. Friction exists between any two materials that contact each other.

Pressure

is the force applied in a perpendicular direction to the surface of an object. Often, pressure is measured in pounds of force exerted per square inch of an object, or PSI.

AIRFLOW AROUND A BODY



THEORIES IN THE PRODUCTION OF LIFT

In order to achieve flight in a machine that is heavier than air, there are several obstacles we must overcome. One of those obstacles, discussed previously, is the resistance to movement called drag. The most challenging obstacle to overcome in aviation, however, is the force of gravity.

A wing moving through air generates the force called lift, also previously discussed. Lift from the wing that is greater than the force of gravity, directed opposite to the direction of gravity, enables an aircraft to fly.

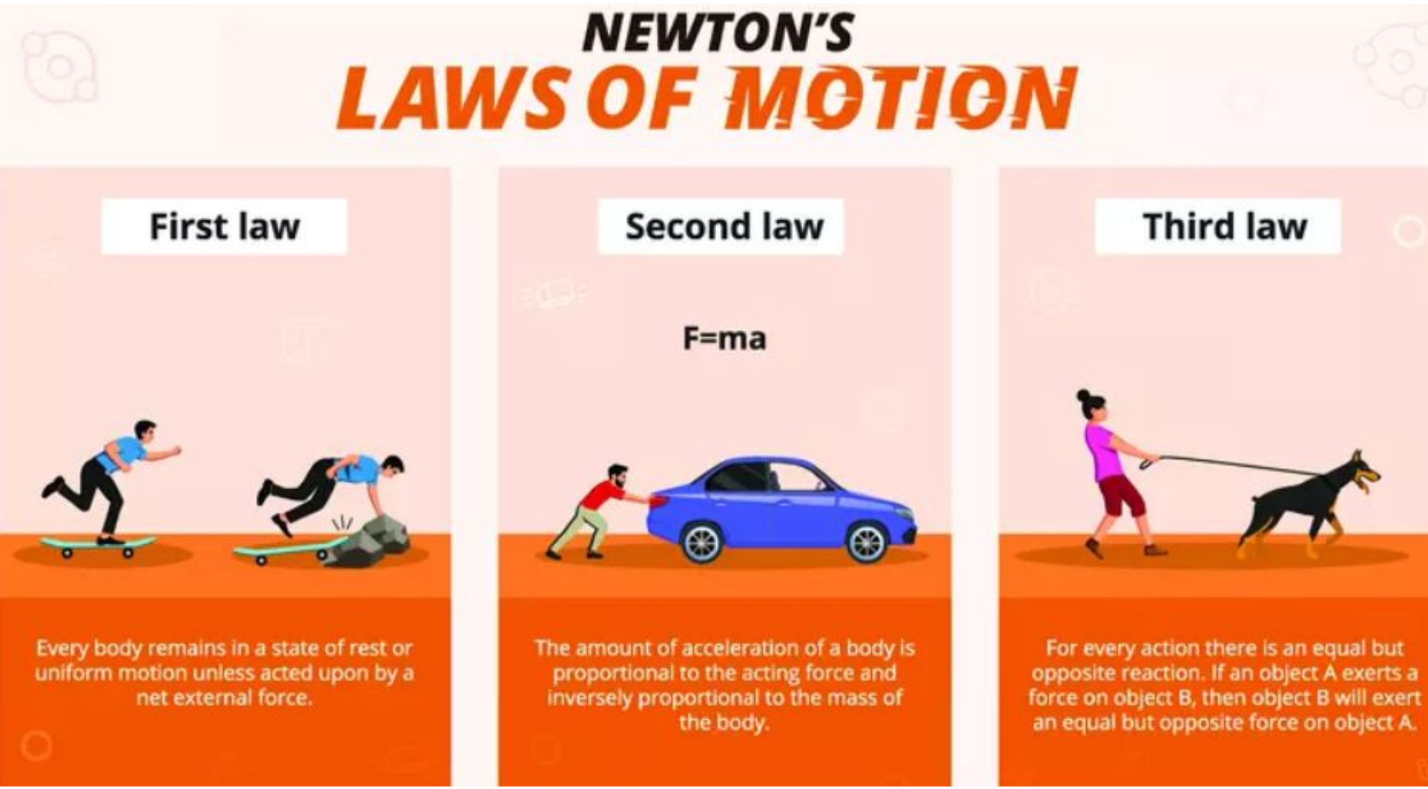
Generating this force called lift is based on some important principles, Newton's basic laws of motion, and Bernoulli's principle of differential pressure.

THEORIES IN PRODUCTION OF LIFT

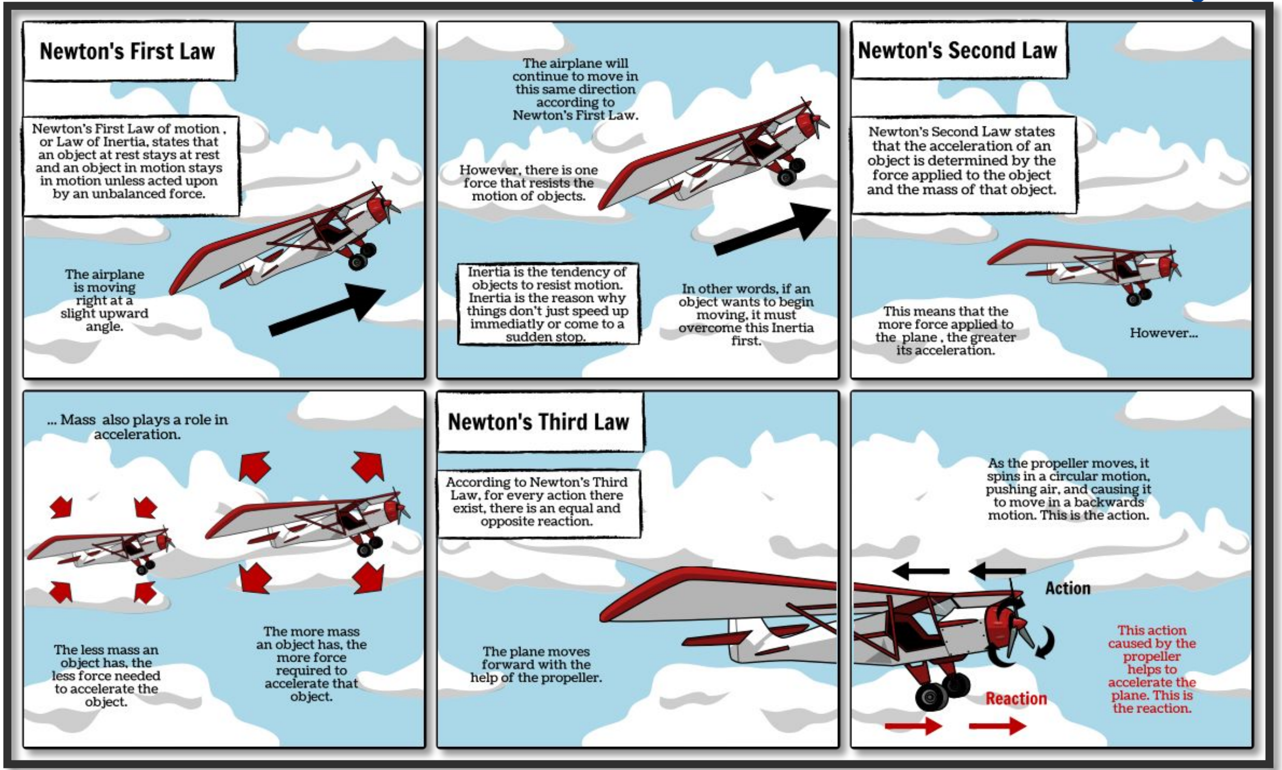
- **Newton's first law** – a body at rest tends to remain at rest, and a body in motion tends to remain at the same speed and in the same direction. For example, an airplane at rest on the ramp will remain at rest unless a force is applied w/c is strong enough to overcome the airplane's inertia

- **Newton's second law** - when a body is acted upon by a constant force, its resulting acceleration is inversely proportional to the mass of the body and is directly proportional to the applied force. This law may be expressed by the formula: Force = mass X acceleration (F = ma)

- **Newton's third law** – for every action there is equal and opposite reaction. This principle applies whenever two things act upon each other, such as the air & propeller or air & the wing.

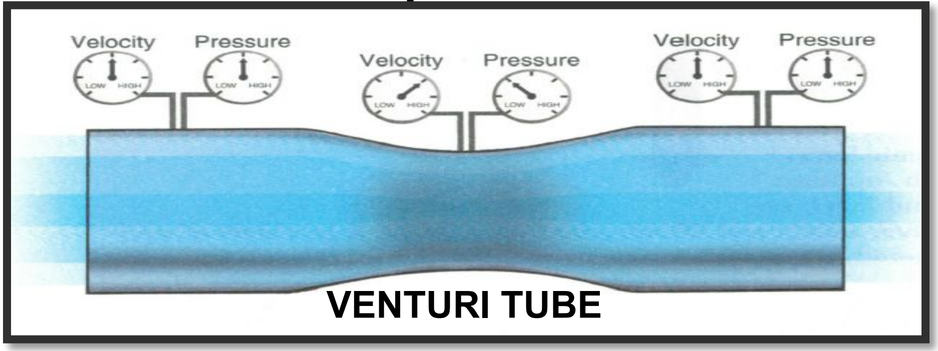

NEWTON'S
LAWS OF MOTION
First law
Every body remains in a state of rest or uniform motion unless acted upon by a net external force.
Second law
F=ma
The amount of acceleration of a body is proportional to the acting force and inversely proportional to the mass of the body.
Third law
For every action there is an equal but opposite reaction. If an object A exerts a force on object B, then object B will exert an equal but opposite force on object A.

THEORIES IN PRODUCTION OF LIFT



THEORIES IN PRODUCTION OF LIFT

Bernoulli's Principle



- Force through a narrow point in a tube is called **"VENTURI"**
- **Air flow top of the wing travels further distance to cover and speeds up and causes an area of low pressure above the wing**
- **Air flow below the wing slows down and has a shorter distance to travel and causes high pressure**

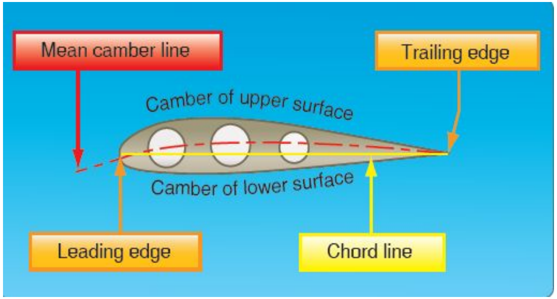
Discovered by Daniel Bernoulli (1700-1782), sates that:

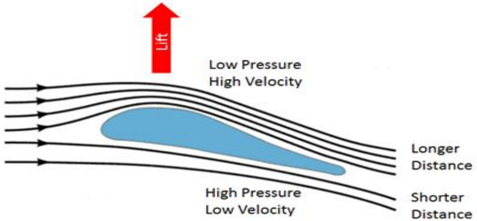
a) "As the velocity of the fluid increases, the pressure in that fluid decreases," and conversely, "as velocity of the fluid decreases, the pressure in the fluid increases

AIRFOIL DESIGN

Airfoil Design

An airfoil is a structure designed to obtain reaction upon its surface from the air through which it moves or that moves past such a structure. Air acts in various ways when submitted to different pressures and velocities



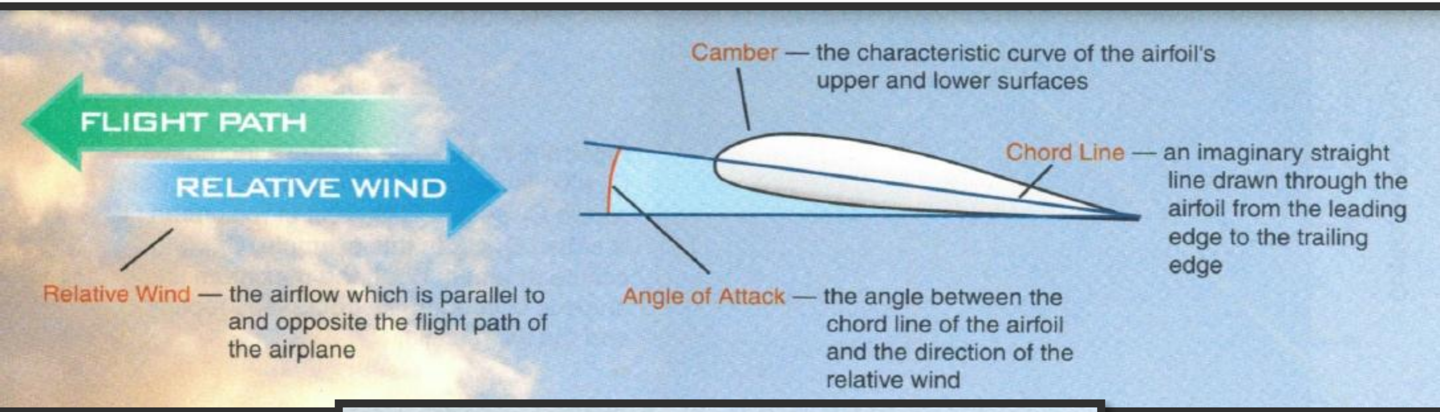


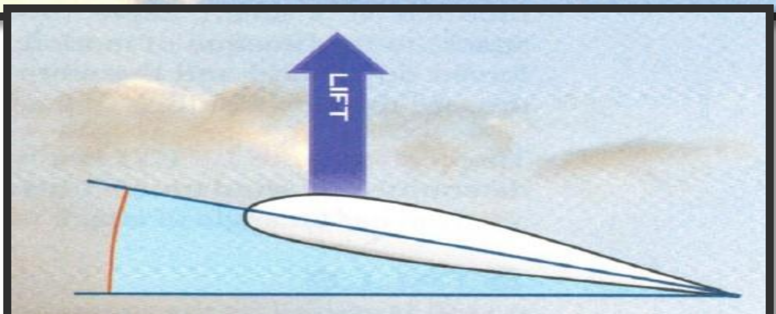
Also known as the "Longer Path" or "Equal Transit" Theory

As an airfoil moves through the air, it alters the airflow around it.

- low pressure is lift- air speeding up over the curved top surface creates a lower pressure than under the wing.

- relative *low pressure at* the top of the wing and a relative high pressure below the wing creates lift which makes airplane fly.





As the angle of attack increases, lift also increases. Notice that lift acts perpendicular to the relative wind, regardless of angle of attack.

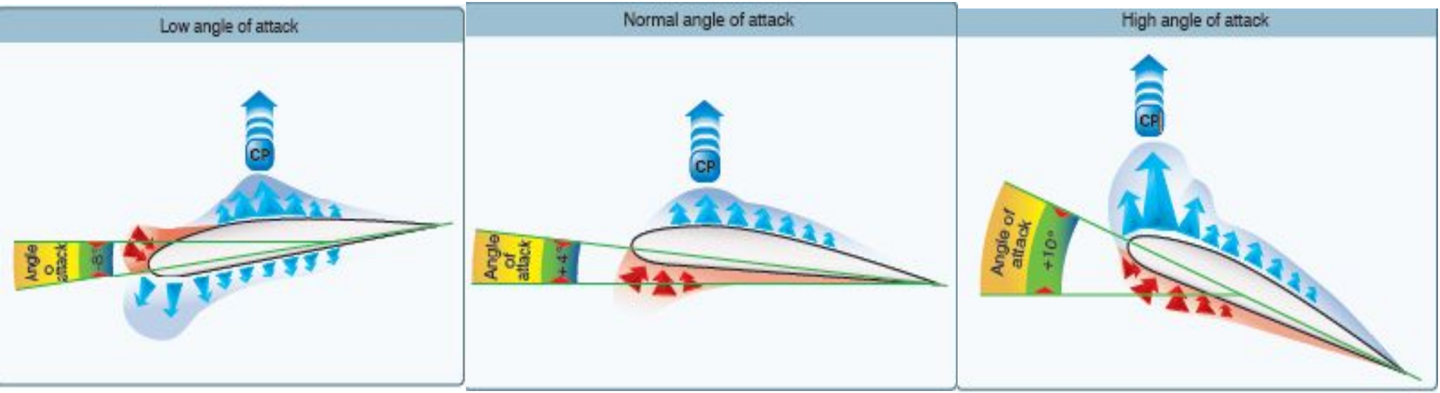
SUB-SONIC

- In subsonic aerodynamics, the theory of lift is based upon the forces generated on a body and a moving gas (air) in which it is immersed. At speeds of approximately 260 knots or less

TWO DIMENSIONAL AIRFOIL

- **Low Pressure Above**
- **High Pressure Below**

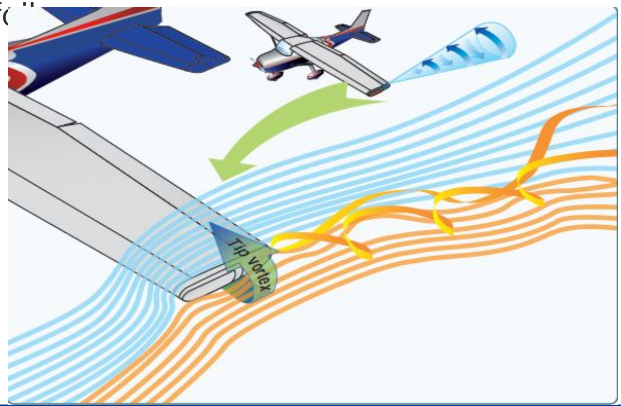
While most of the lift is produced by these two dimensions, To this point, the discussion has centered on the flow across the upper and lower surfaces of an airfoil.

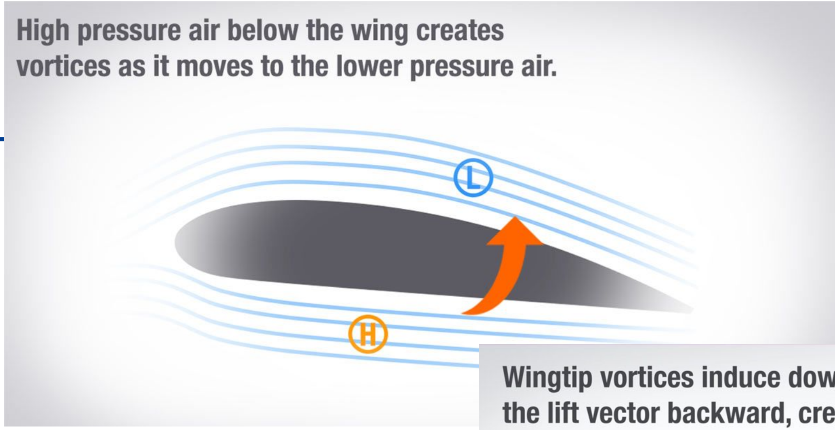


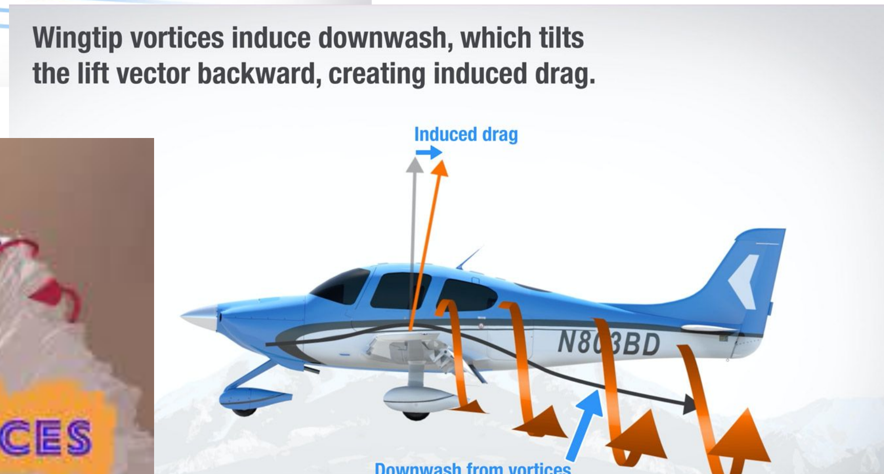
THIRD DIMENSION

A Third Dimension

- The high-pressure area on the bottom of an airfoil pushes around the tip to the low-pressure area on the top.
- This action creates a rotating flow called a tip vortex. The vortex flows behind the airfoil creating a downwash that extends back to the trailing edge of the airfoil. This downwash results in an overall reduction in lift for the affected portion of the airf

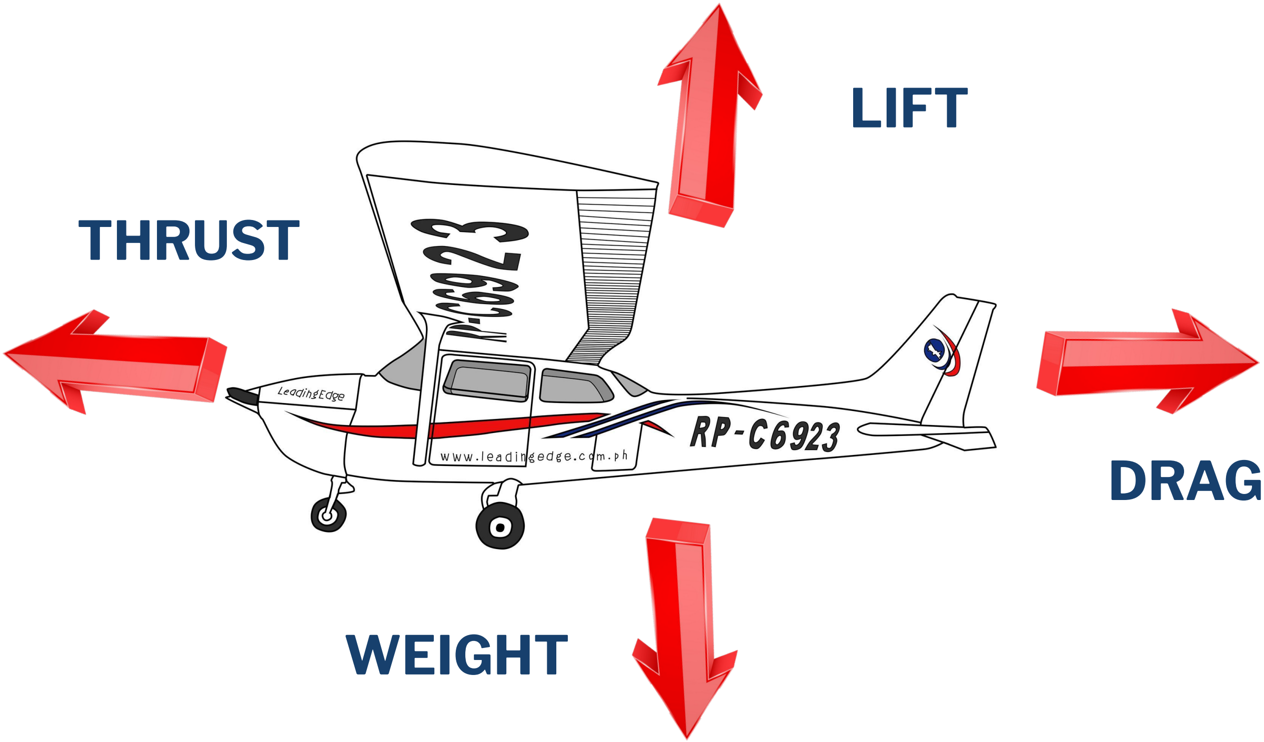



High pressure air below the wing creates vortices as it moves to the lower pressure air.
L
H

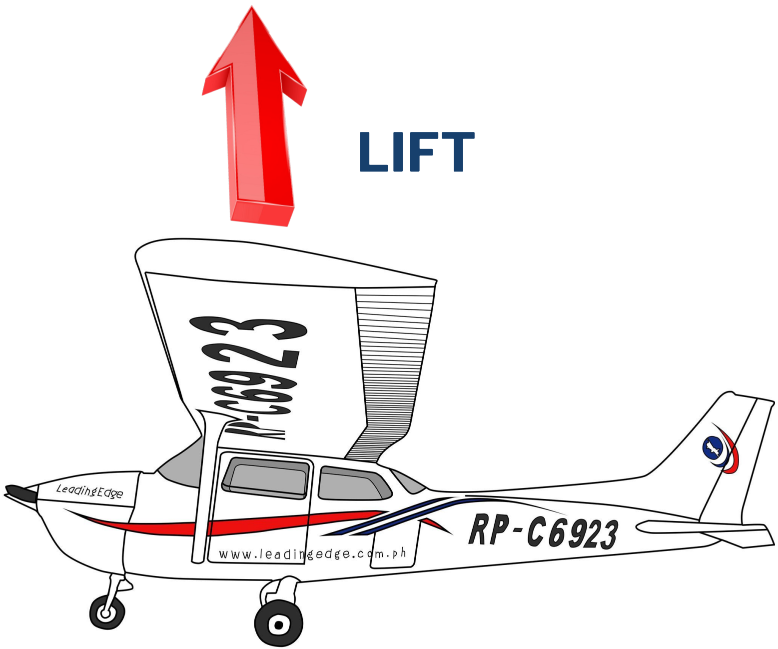

Wingtip vortices induce downwash, which tilts the lift vector backward, creating induced drag.
Induced drag
N803BD
Downwash from vortices


WINGTIP VORTICES

FOUR FORCES OF FLIGHT



FOUR FORCES OF FLIGHT



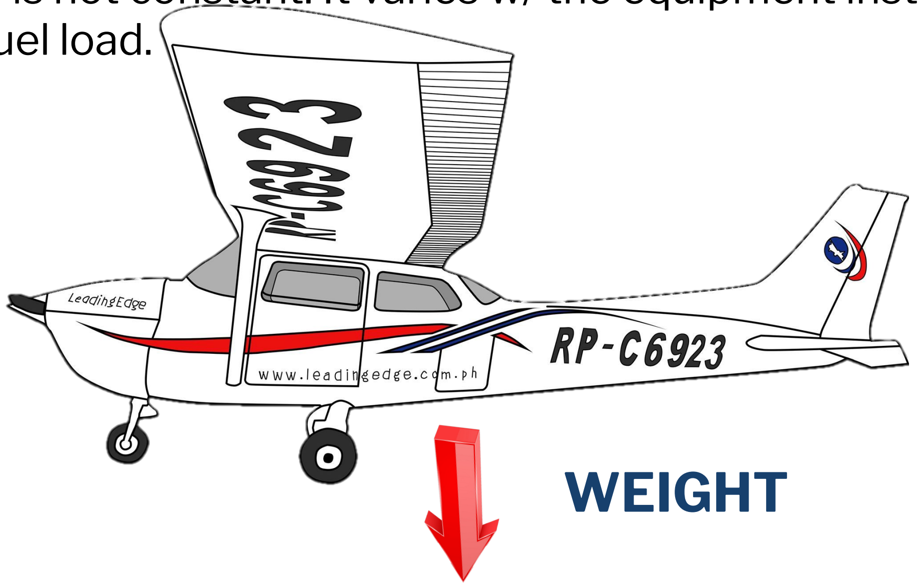
Lift – differential pressure between the upper and lower camber

Kinds of Lift:

- **Dynamic** – derived by the impact of pressures on the lower camber which is only about 25% of the total lifting force.
- **Induce** – produced by the low pressure above the upper camber of an airfoil.

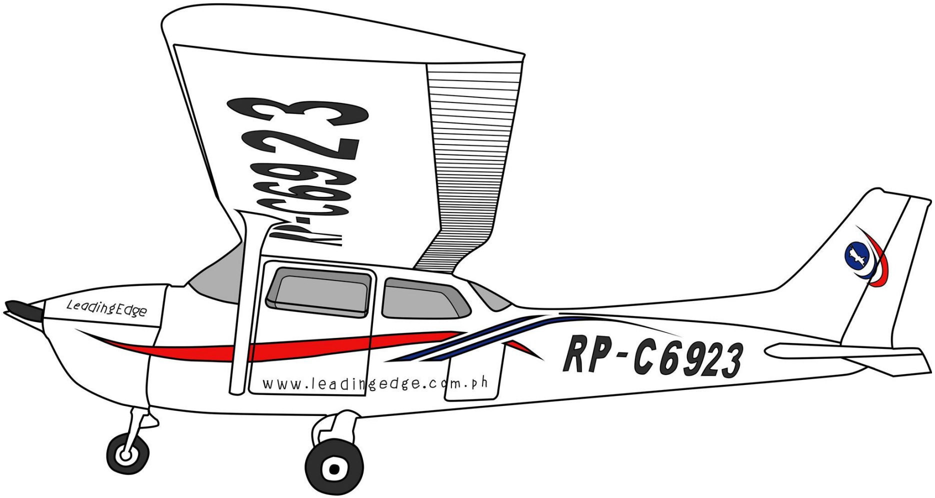
FOUR FORCES OF FLIGHT

Weight is the force of gravity which acts vertically through the center of the airplane toward the center of the earth. The weight of the of the airplane is not constant. It varies w/ the equipment installed, pax, cargo and fuel load.



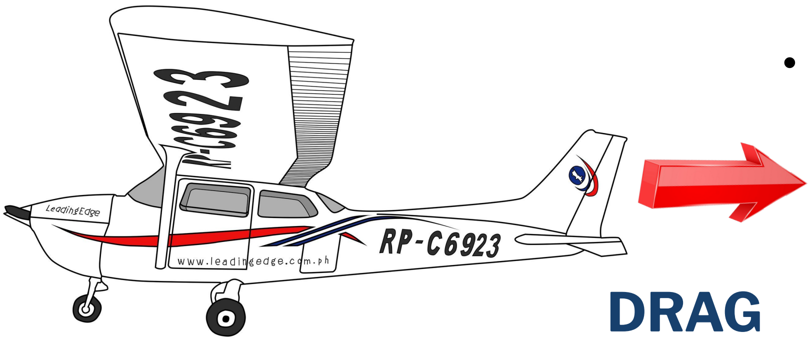
FOUR FORCES OF FLIGHT

THRUST



Thrust is the forward-acting force w/c opposes drag and propels the airplane. In most general aviation airplanes, this forces is provided when the engine turns the propeller.

FOUR FORCES OF FLIGHT

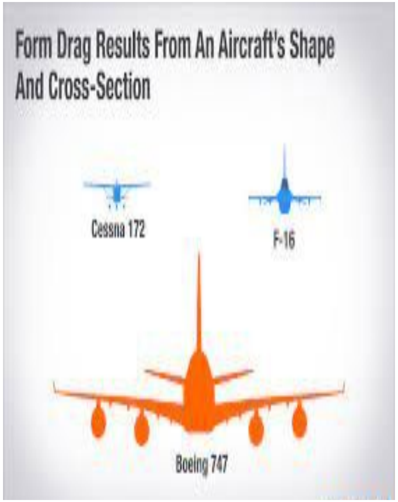


DRAG

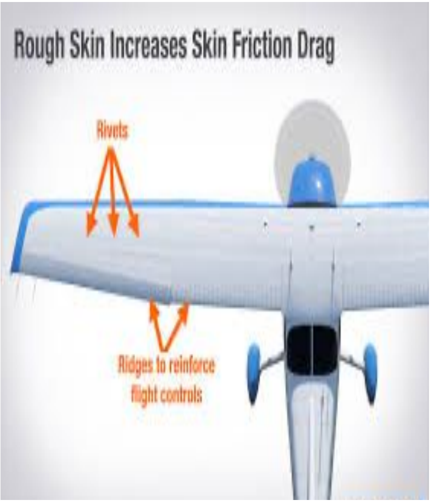
- **Drag acts** in opposition to the direction of flight, opposes the forward-acting force of thrust, and limits the forward speed of the airplane.
- **Parasite drag** – is caused by any aircraft surface which deflects or interferes w/ the smooth airflow around the airplane.
  - Interference drag
  - Form drag
  - Skin friction
- **Induced drag** – is generated by the airflow circulation around the wing as it creates lift.

TYPES OF PARASITE DRAG

FORM DRAG

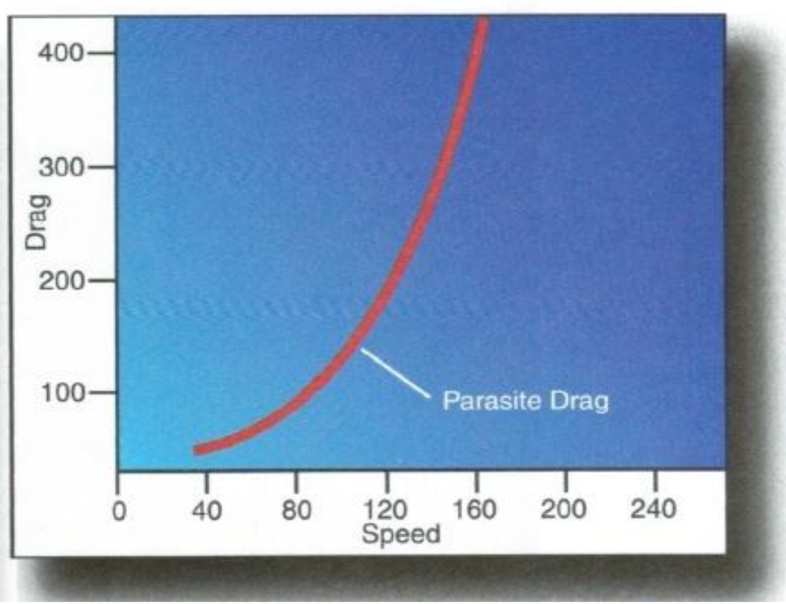


SKIN FRICTION



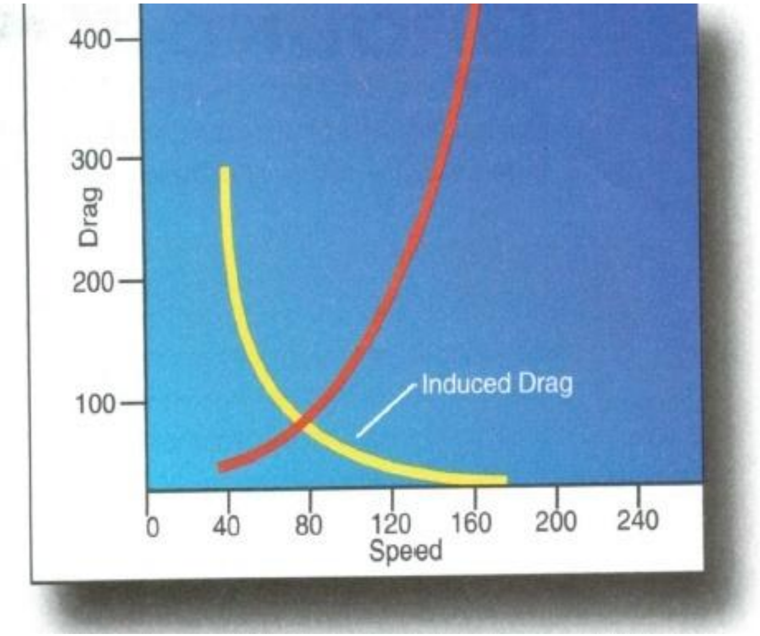
INTERFERENCE DRAG

PARASITE DRAG



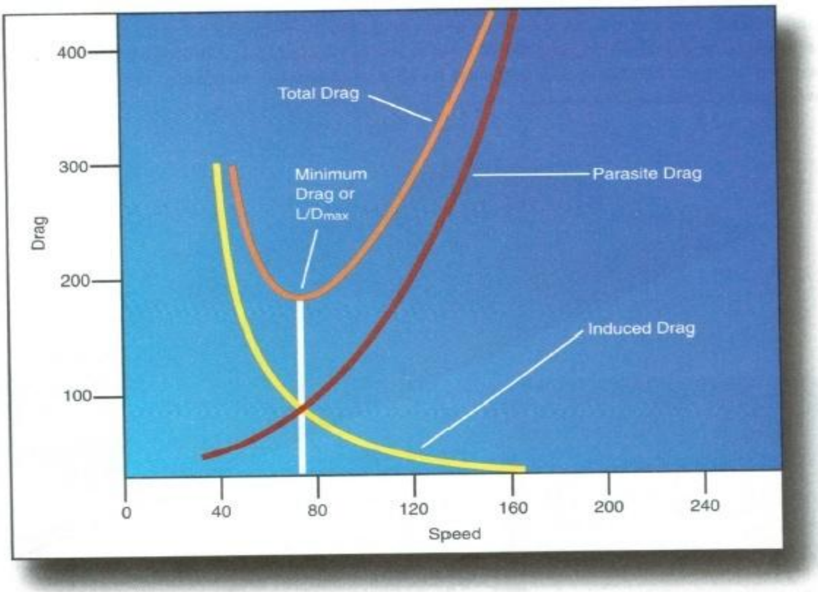
If airspeed is doubled, **parasite drag** increases fourfold. This is the same formula that applies to lift. Because of its rapid increase with increasing airspeed, parasite drag is predominant at high speeds. At low speeds, near a stall, parasite drag at its low point.

INDUCED DRAG



Induced drag is inversely proportional to the square of the speed. If speed is decreased by half, induced drag increases fourfold. It is the major cause of drag at reduced speeds near the stall; but, as speed increases, induced drag decreases.

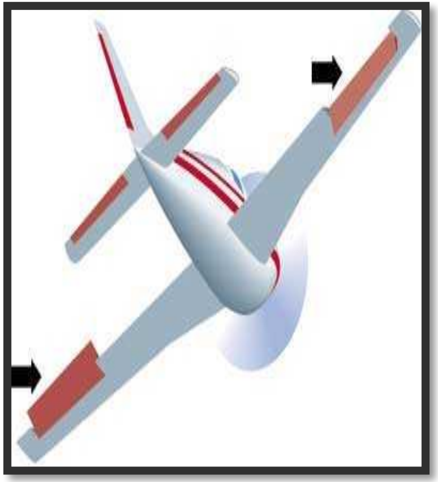
TOTAL DRAG



The low point on the total drag curve shows the airspeed at which drag is minimized. This point, where the lift-to drag ratio is greatest, is referred to as **L/D_{MAX}.** At this speed, the total lift capacity of the airplane, when compared to the total drag of the airplane, is most favorable. This is important in airplane performance.

PRIMARY FLIGHT CONTROLS

Ailerons – extend from about the midpoint of each wing outward to the tip and move in opposite direction to create aerodynamic force that causes the airplane to turn

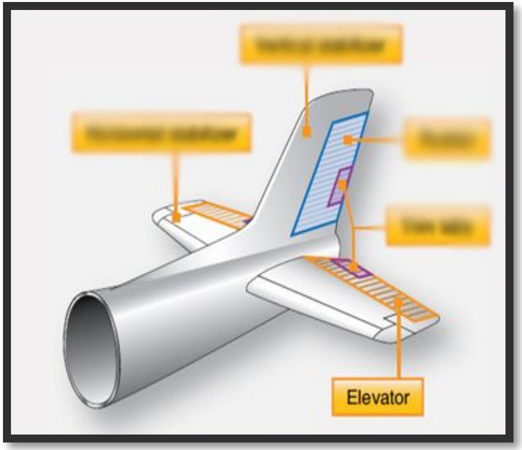


PRIMARY FLIGHT CONTROLS

Elevator– is controlled by the control wheel through a system of cables, pulleys, and other connecting devices.

-attached at the rear of the horizontal stabilizer

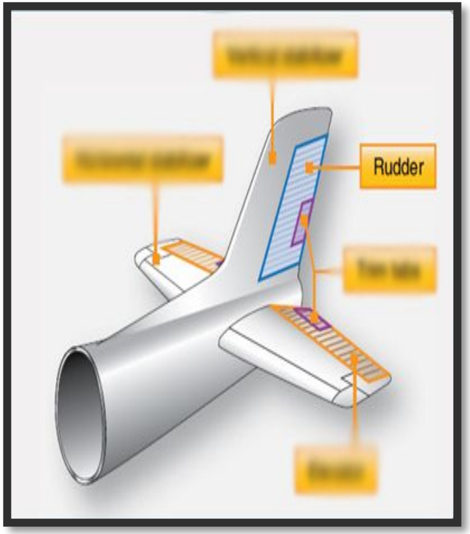
-Controls the ATTITTUDE (pitching) nose up or down of the aircraft



PRIMARY FLIGHT CONTROLS

Rudder – You operate the rudder with your feet, using pedals located in the cockpit. When you press the left rudder pedal, connecting cables move the rudder to the left which causes the airplane's nose to move to the left. Pressing the right pedal moves the rudder and nose to the right.

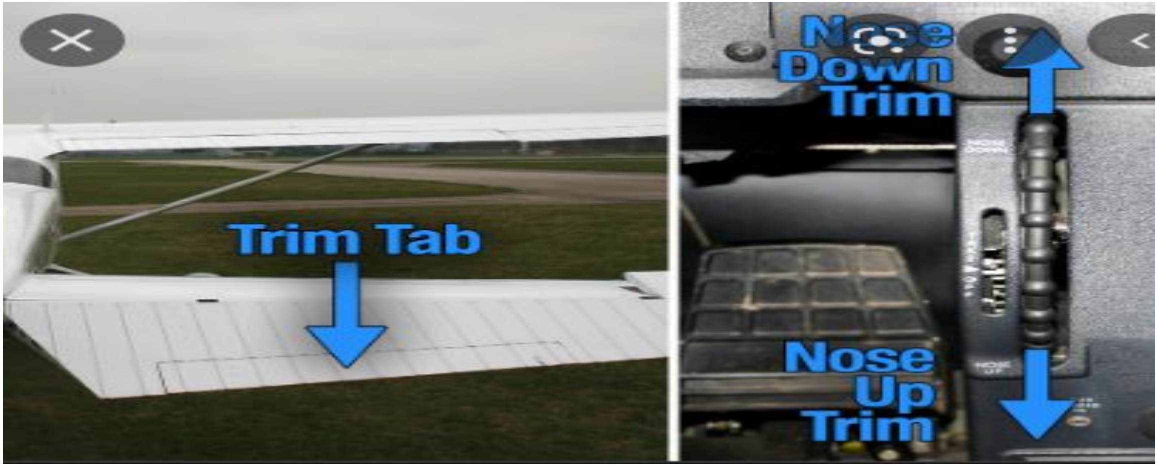
- steers the airplane on the ground when **taxiing (rudder pedals)**



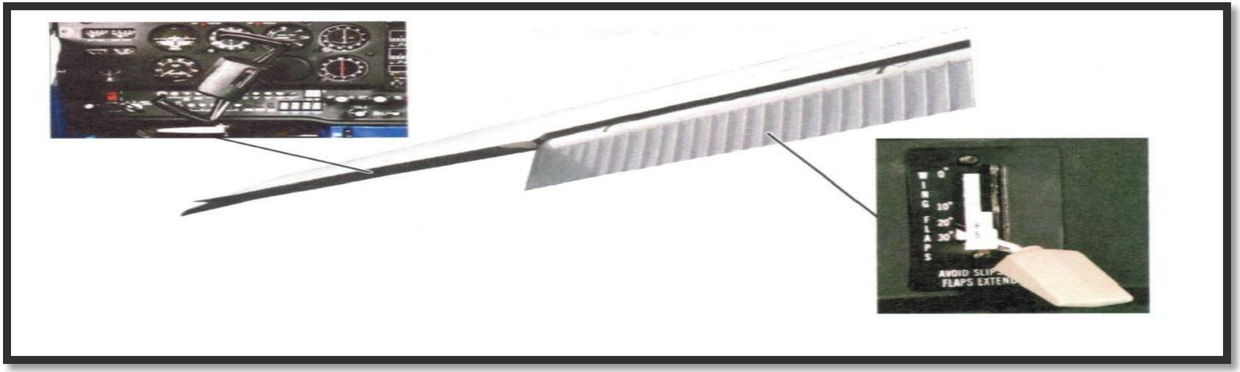
SECONDARY FLIGHT CONTROLS

- The **TRIM TAB** is used to 'trim' the aircraft and allow it to be flown at various attitudes with minimal pilot control force.

- The elevator trim system consists of a cockpit-mounted wheel and position indicator connected to a pair of cables which extend, through a series of pulleys, to the elevator trim tab actuator



SECONDARY FLIGHT CONTROLS

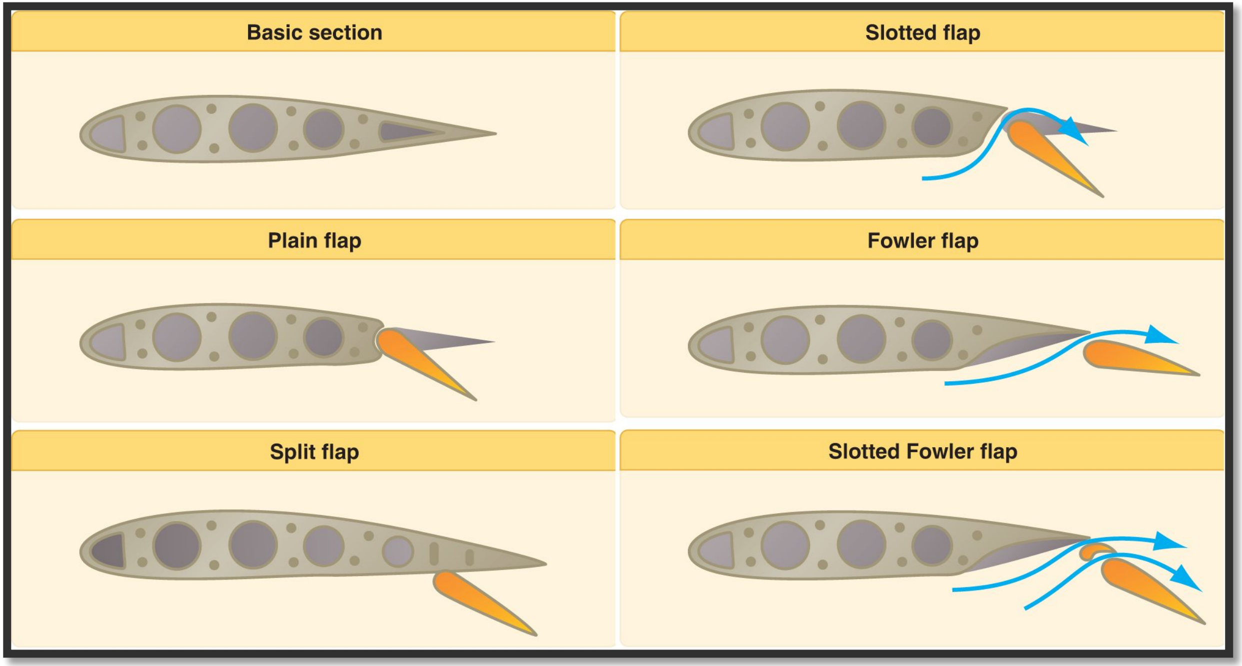


Flaps – extend outward from the fuselage to the midpoint of the each wing. The flaps are normally flush with the wing's surface during cruising flight. When extended, the flaps move simultaneously downward to increase the lifting force of the wing for take-off and landing.

- Main function in landing is **increasing rate of descent without increasing airspeed**

- *Lifting device and breaking device*

TYPES OF FLAPS

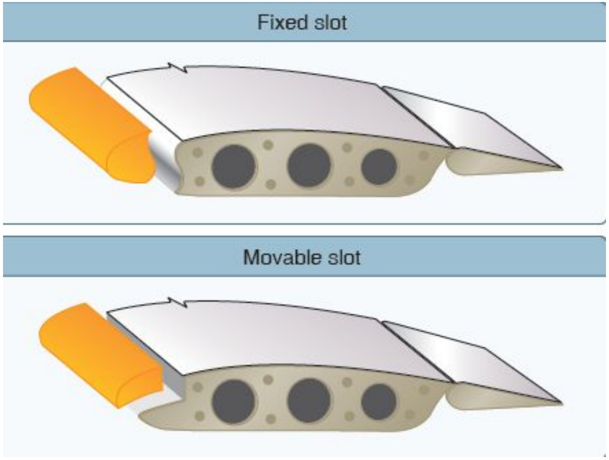


SLATS

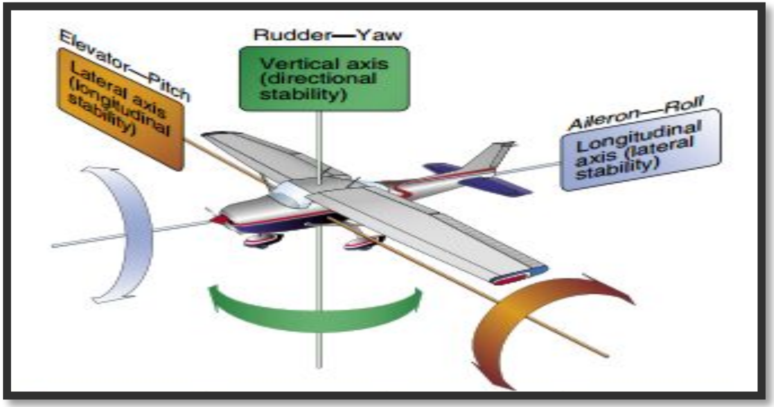
Leading Edge Devices

- High-lift devices also can be applied to the leading edge of the airfoil. The most common types are fixed slots, movable slats, leading edge flaps, and cuffs.

- Fixed slots direct airflow to the upper wing surface and delay airflow separation at higher angles of attack.

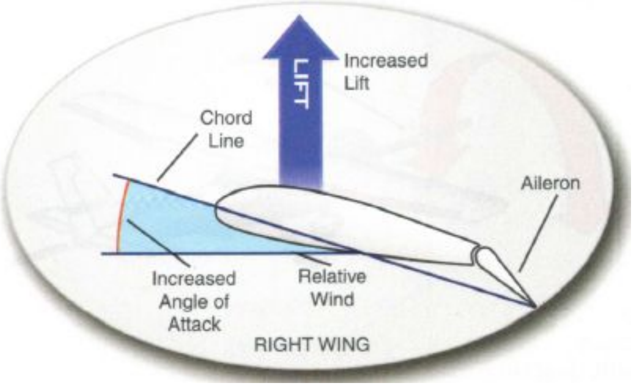


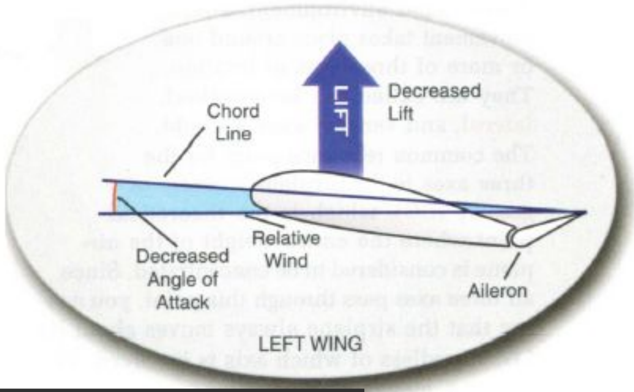
FLYING CONTROLS



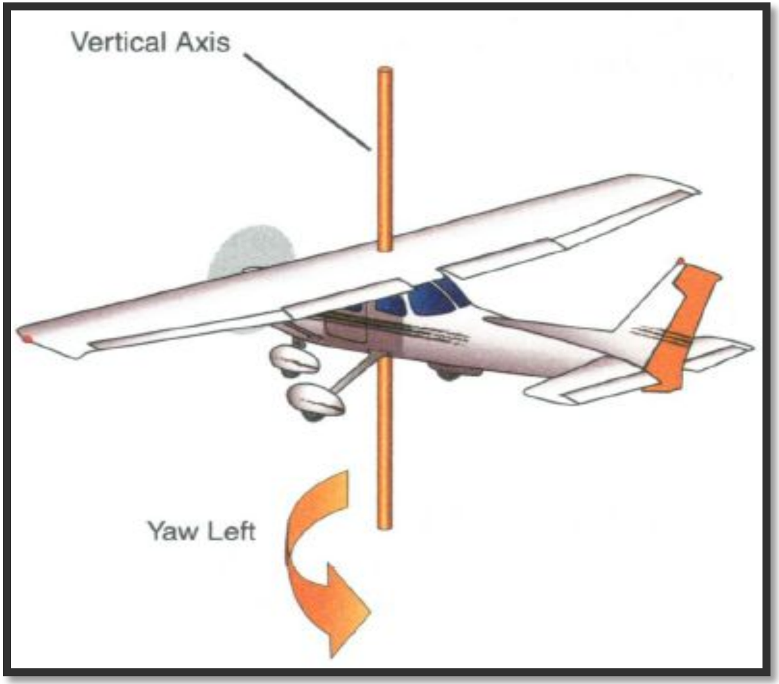
Cockpit Control	Outside Control Surfaces	Motion	Axes
Yoke	Aileron	Roll	Longitudinal
Rudder	Tail Rudder	Yaw	Vertical
Yoke	Elevator	Pitch	Lateral

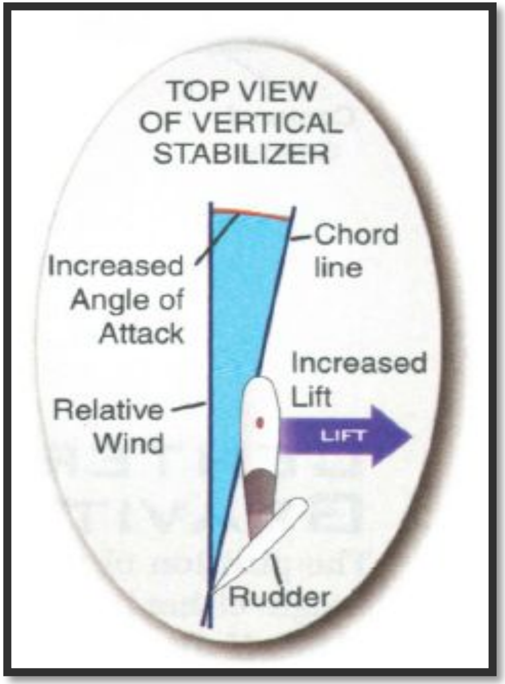
LONGITUDINAL AXIS



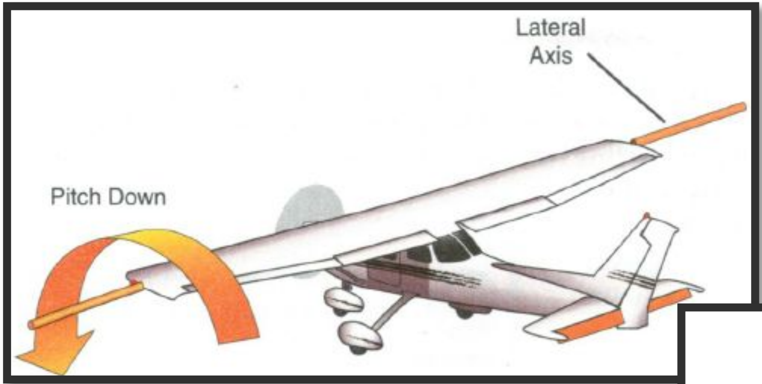


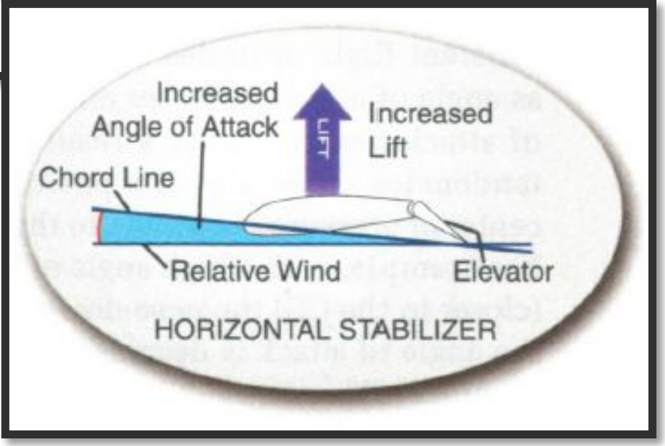
VERTICAL AXIS





LATERAL AXIS





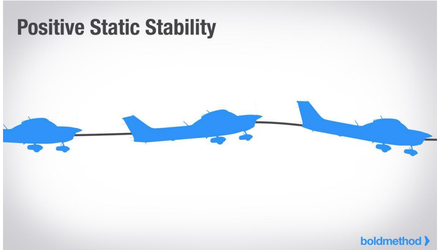
STABILITY

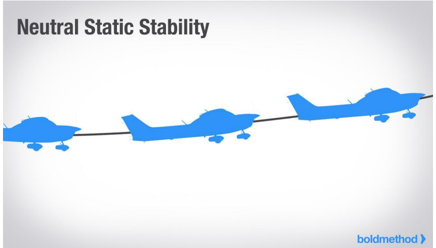
Stability is the characteristic of an airplane in flight that causes it to return to a condition of equilibrium, or steady flight, after it is disturbed.

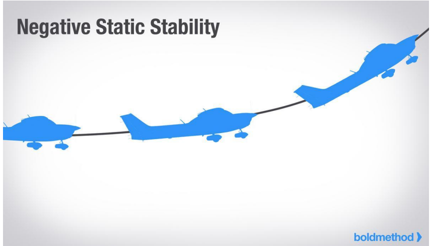
The initial tendency to return to the position form w/c it was displaced is termed positive static stability. However, since the aircraft doesn't immediately return to the original position, but instead does so over a period of time through a series of successively smaller oscillations, the aircraft also displays positive dynamic stability.

STATIC STABILITY

Static Stability Static stability refers to the initial tendency, or direction of movement, back to equilibrium. In aviation, it refers to the aircraft's initial response when disturbed from a given pitch, yaw, or bank.







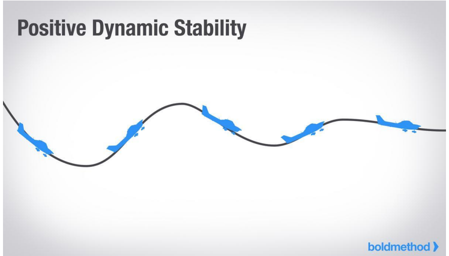
- *Positive static stability* — the initial tendency of the aircraft to return to the original state of equilibrium after being disturbed.

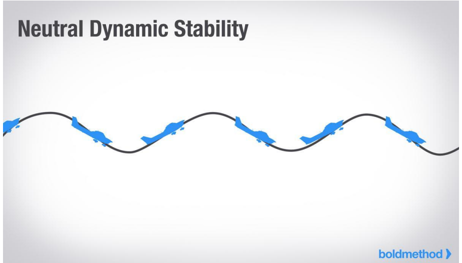
- *Neutral static stability* — the initial tendency of the aircraft to remain in a new condition after its equilibrium has been disturbed.

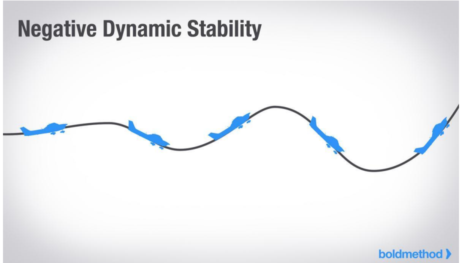
- *Negative static stability* — the initial tendency of the aircraft to continue away from the original state of equilibrium after being disturbed.

DYNAMIC STABILITY

Dynamic stability refers to the aircraft response over time when disturbed from a given pitch, yaw, or bank. This type of stability also has three subtypes:







• *Positive dynamic stability* — over time, the motion of the displaced object decreases in amplitude and, because it is positive, the object displaced returns toward the equilibrium state.

• *Neutral dynamic stability* — once displaced, the displaced object neither decreases nor increases in amplitude. A worn automobile shock absorber exhibits this tendency.

• *Negative dynamic stability* — over time, the motion of the displaced object increases an becomes more divergent.

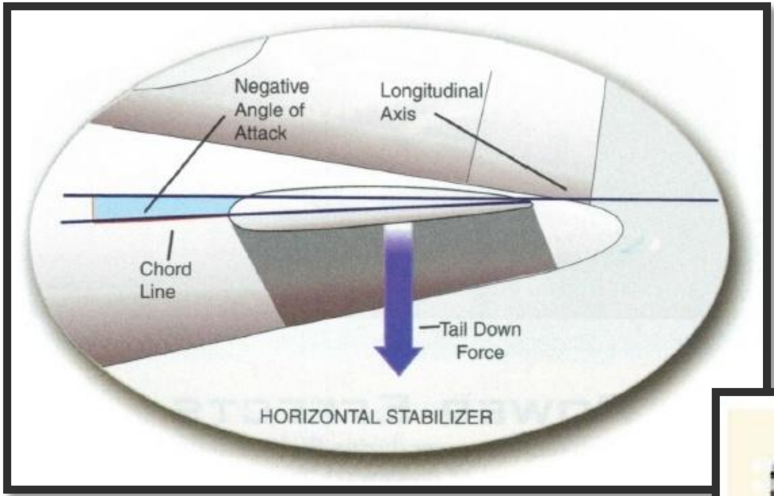
Stability is the inherent quality of an aircraft to correct for conditions that may disturb its equilibrium and to return to or to continue on the original flight path.

Longitudinal stability – of an airplane involves the pitching motion or tendency of the airplane to move about the lateral axis.

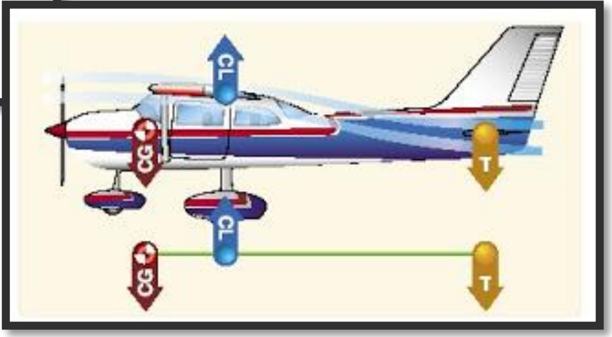
Lateral stability – stability about an airplane's longitudinal axis. This tendency to resist lateral, or roll movement.

Directional stability – stability about the vertical axis.

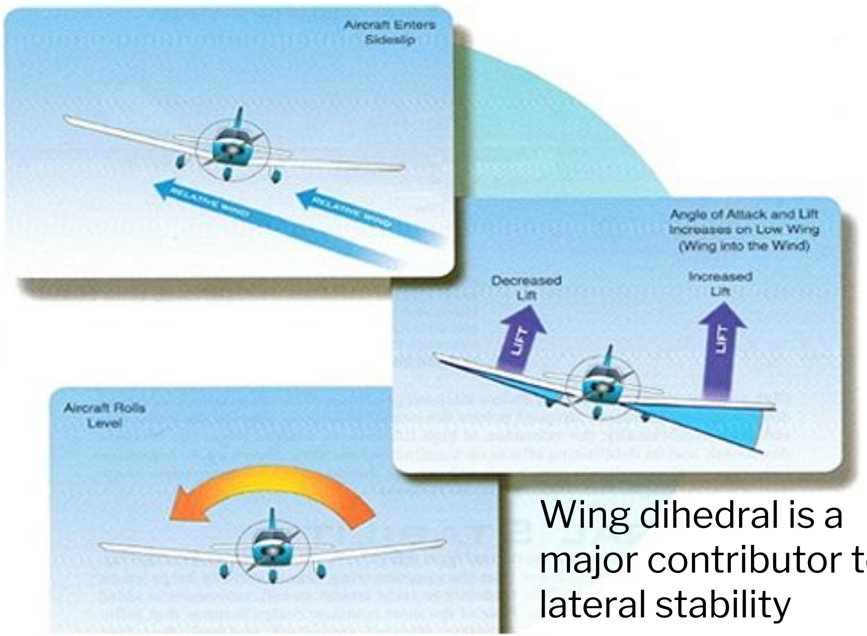
LONGITUDINAL STABILITY



To Provide Longitudinal Stability, Most single engine airplanes are designed to have a negative angle of attack on the horizontal stabilizer to compensate for the nose heaviness



LATERAL STABILITY



Wing dihedral is a major contributor to lateral stability

To provide Lateral Stability, airplane wings have Dihedral Angle. Dihedral is an upward angle of the airplane's wing with respect to the horizontal.

DIRECTIONAL STABILITY

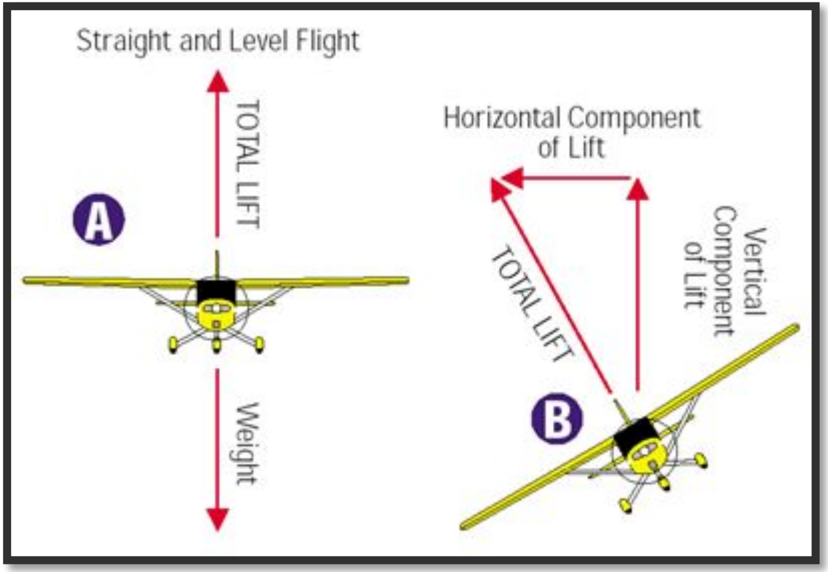
An airplane must have more surface area behind the CG than it has infront of it. When an airplane enters a sideslip, the greater the surface area behind the CG helps keep the airplane aligned with the relative wind.

The primary contributor to directional stability is the vertical tail w/c causes an airplane in flight to act much like a weather vane.

HOW AN AIRPLANE TURNS

How an Airplane Turns

It's the bank that makes the airplane turn. Turns are made by inclining the lift of the wings.



The resultant of the horizontal component of lift (centripetal force) and the vertical component of lift is the total lift provided by the wings. When the total lift equals total weight, the aircraft will neither gain nor lose altitude while in the turn.

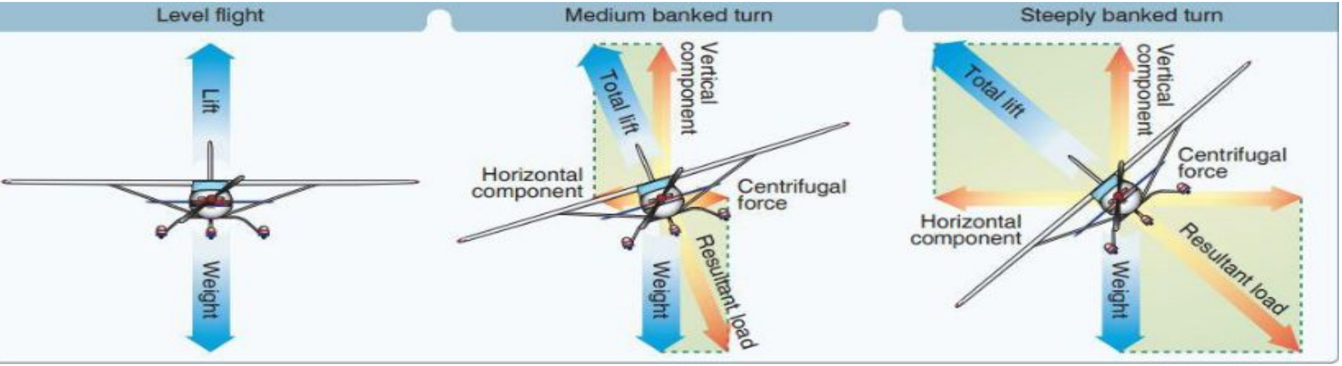
LOAD FACTOR AND FLIGHT MANEUVER

- **Load Factors and Flight Maneuvers**

Critical load factors apply to all flight maneuvers except unaccelerated straight flight where a load factor of 1 G is always present. Certain maneuvers considered in this section are known to involve relatively high load factors. Full application of pitch, roll, or yaw controls should be confined to speeds below the maneuvering speed.

- **TURNS**

Increased load factors are a characteristic of all banked turns. As noted in the section on load factors in steep turns, load factors become significant to both flight performance and load on wing structure as the bank increases beyond approximately 45°.



LOAD FACTOR

The load factor imposed on an airplane will increase as the angle of bank is increased.

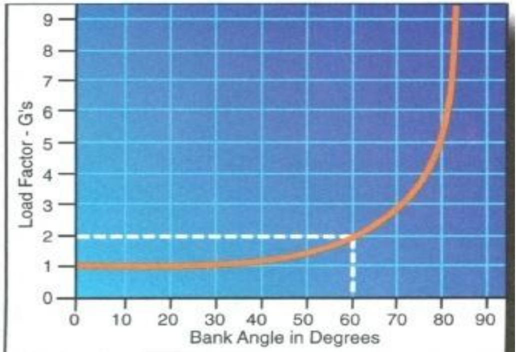


Figure 3-68. During constant altitude turns, the relationship between load factor, or G's, and bank angle is the same for all airplanes. For example, with a 60° bank, two G's are required to maintain level flight. This means the airplane's wings must support twice the weight of the airplane and its contents, although the actual weight of the airplane does not increase.

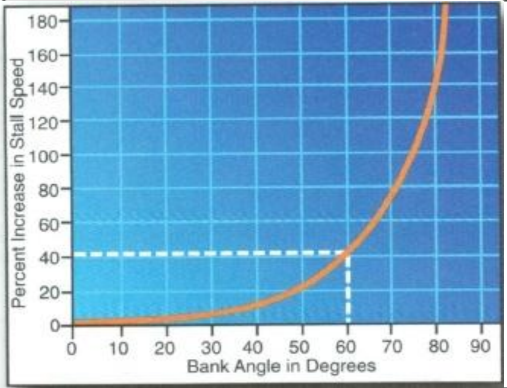


Figure 3-69. If you attempt to maintain altitude during a turn by increasing the angle of attack, the stall speed increases as the angle of bank increases. The percent of increase in stall speed is fairly moderate with shallow bank angles — less than 45°. However, once you increase the bank angle beyond 45°, the percent of increase in the stall speed rises rapidly. For example, in a 60°, constant-altitude bank, the stall speed increases by approximately 41%; a 75° bank increases stall speed by about 100%.

Increasing the load factor will cause an airplane to stall at a higher speed.

LEFT TURNING TENDENCY

- Tendency to turn left at **high power settings at low speeds** (take-off).

- Engine is angled a little to the right and the back stabilizer is angled a little to the right. Left wing has somewhat larger angle of attack than the right. Trim tab on the right. All are designed to counteract left turning tendencies.

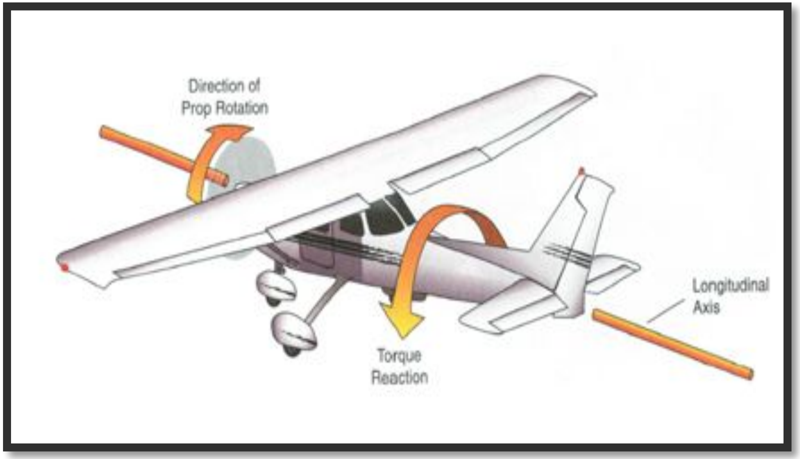
- When flying slower than cruise or using lots of power (climb) you need to help out by applying right rudder.

- Solution is to apply just enough **RUDDER PRESSURE-** on the side you are leaning toward.

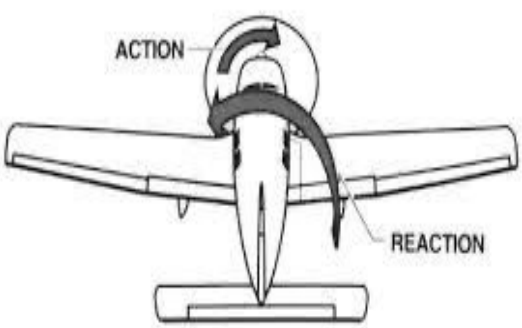
- **INCLINOMETER-** ball is free to move inside the tube and reacts the same way your body does to an unbalanced turn.

- **"STEP ON THE BALL".**

TORQUE EFFECT

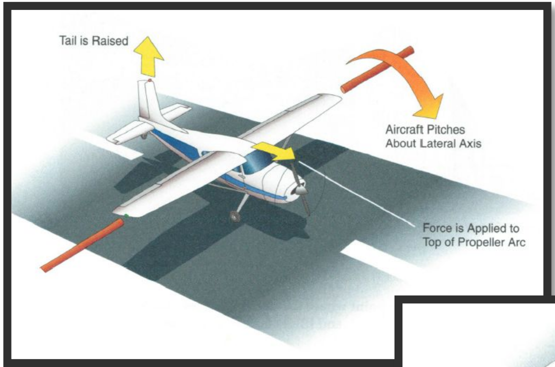


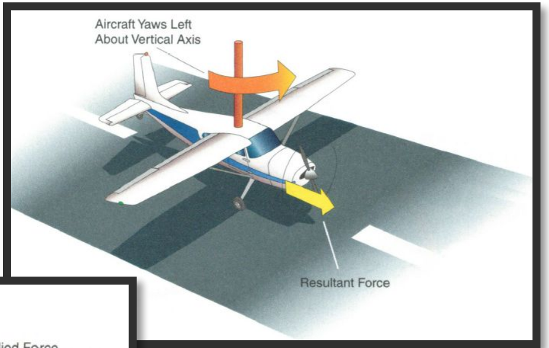
Torque is the opposite reaction created by the turning propeller.

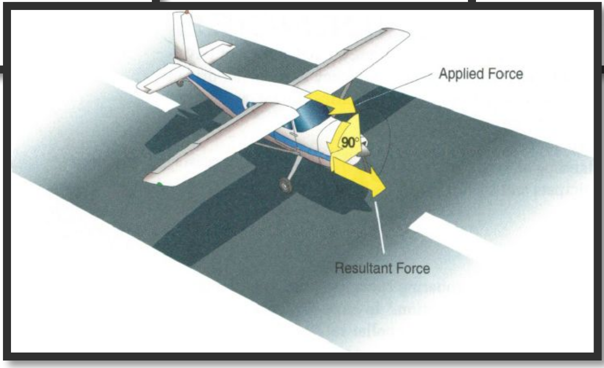


LEFT TURNING TENDENCY

GYROSCOPIC EFFECT

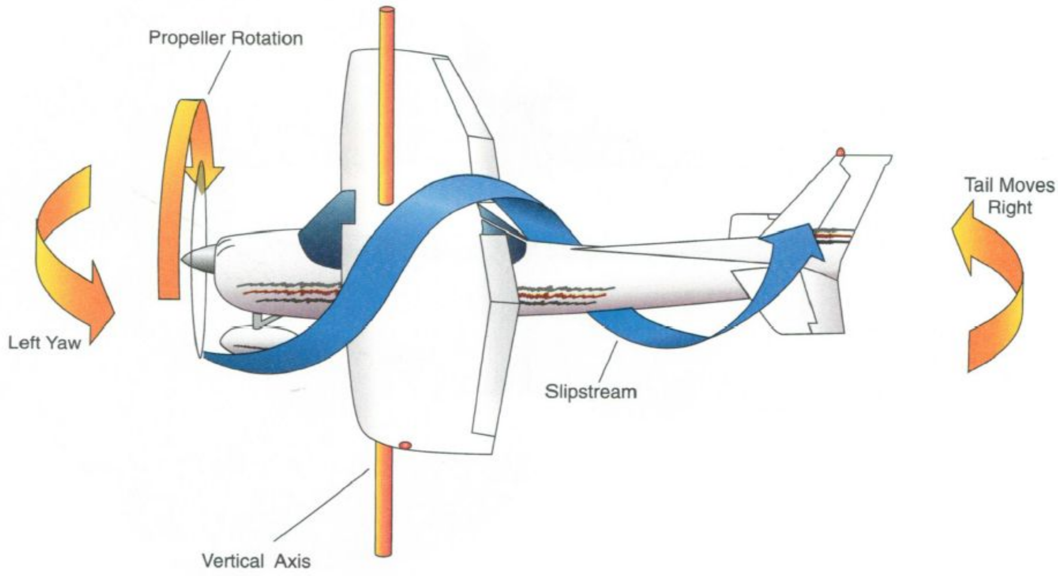






LEFT TURNING TENDENCY

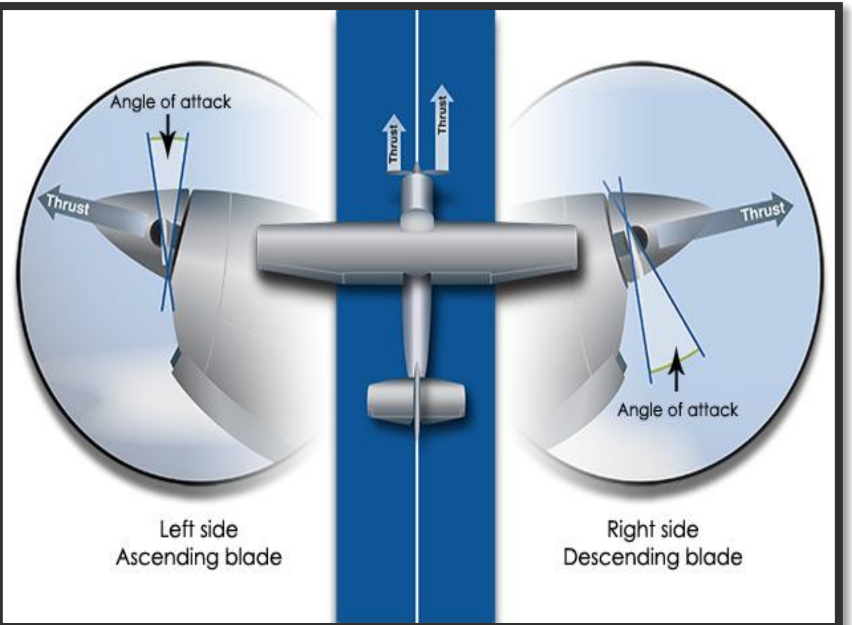
SLIPSTREAM-propeller rotates the air, as well as shipping back along the fuselage



LEFT TURNING TENDENCY

P- FACTOR

Asymmetrical thrust



occurs when an airplane is flown at a high angle of attack.

This causes uneven angles of attack between the ascending and descending propeller blades.

Consequently, less thrust is produced from the ascending blade on the left than from the descending blade on the right.

This produces a tendency for the airplane to yaw to the left.

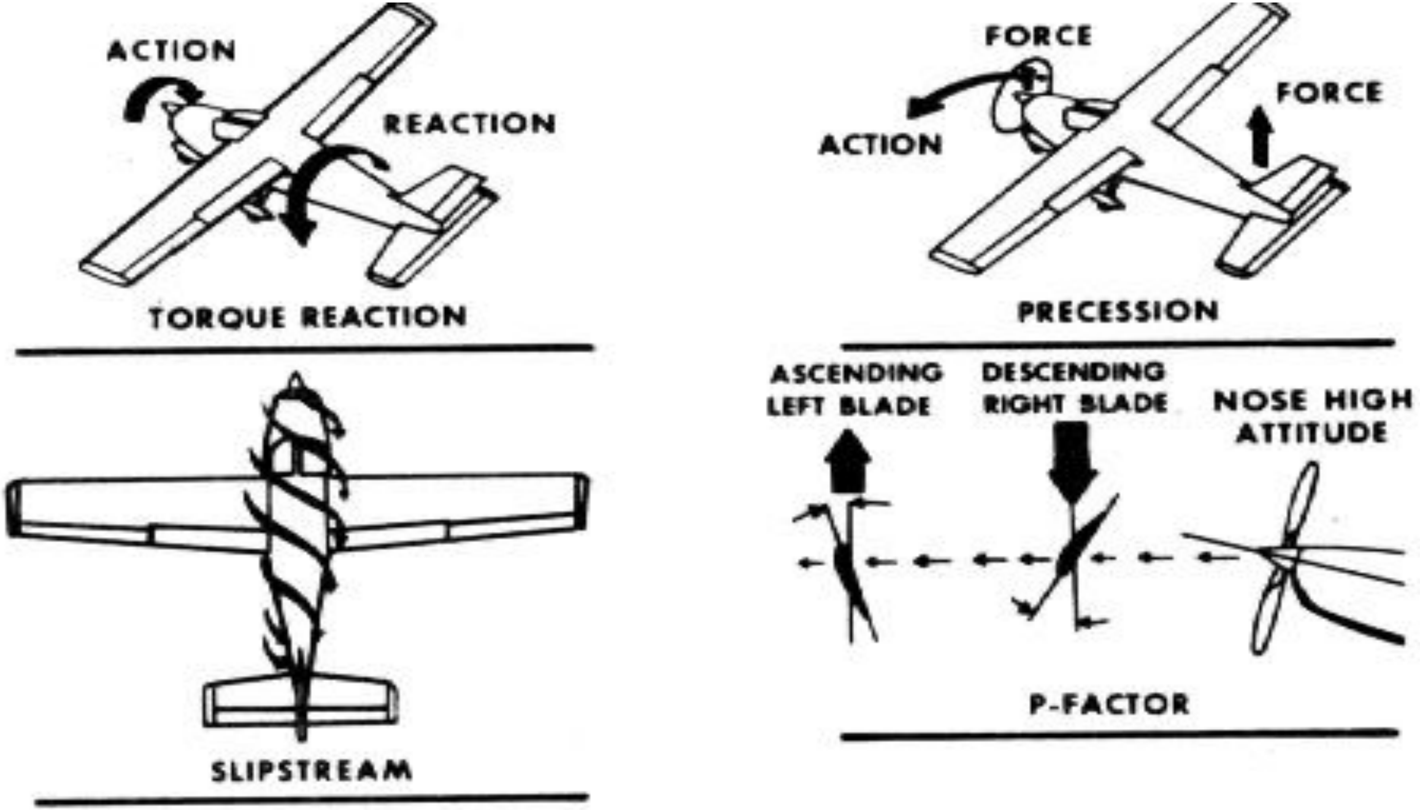
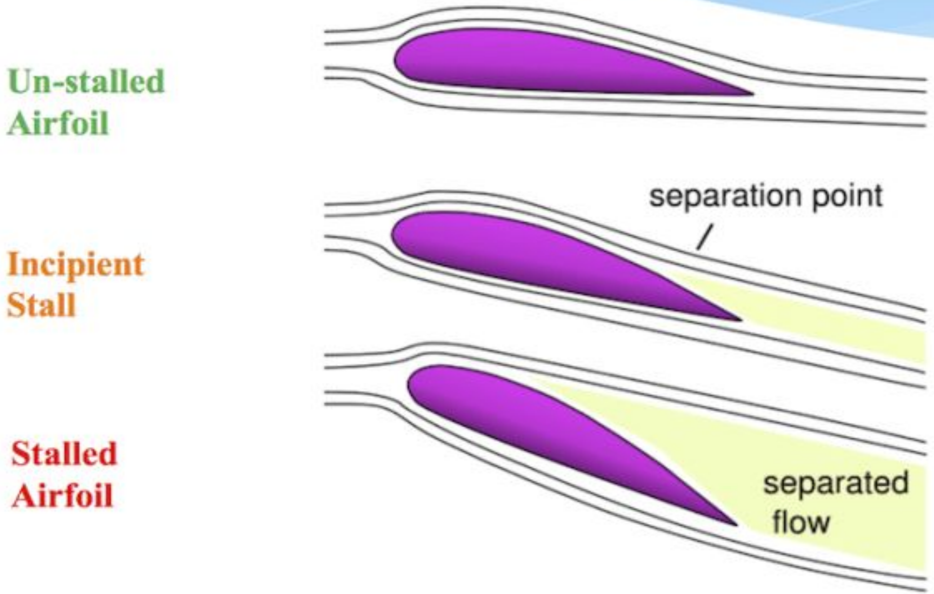
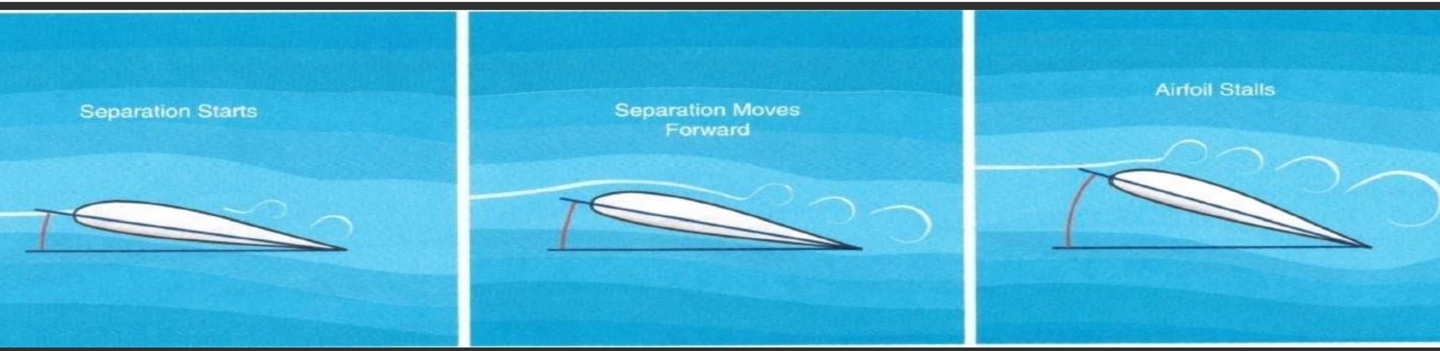


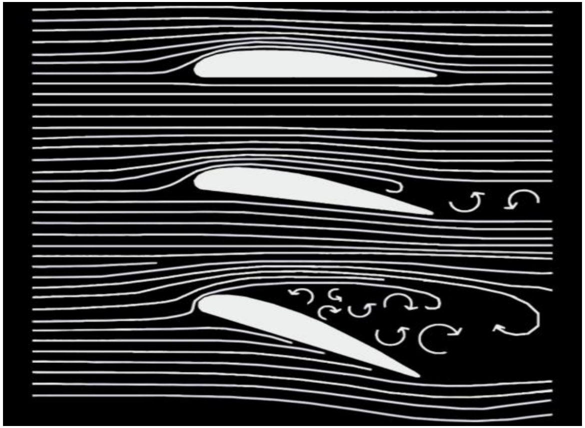
Figure 3–6 Left Turning Tendencies

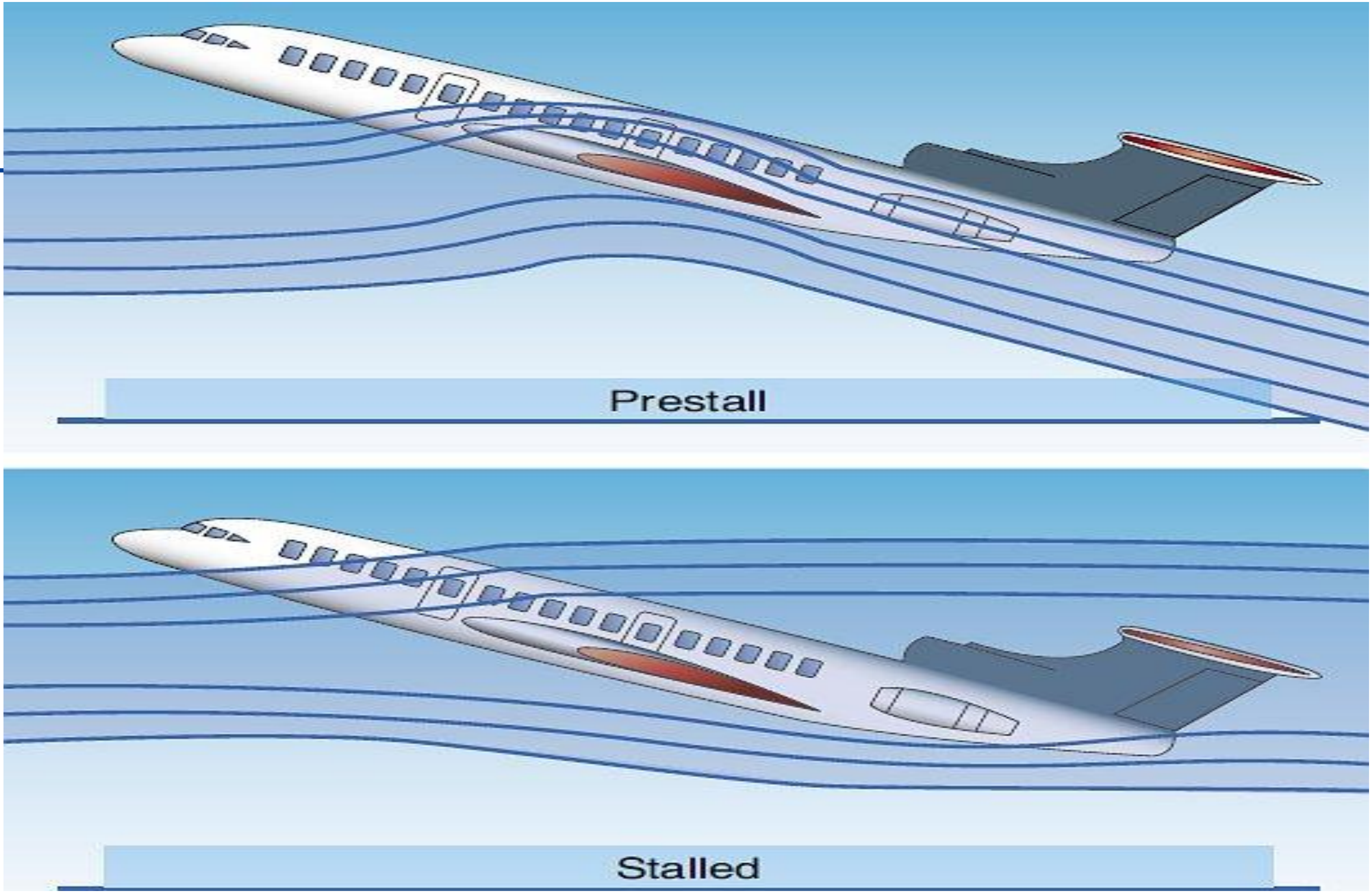
Airfoil Stalls



STALL - The Wing is no longer generating lift


Separation Starts
Separation Moves Forward
Airfoil Stalls




Prestall
Stalled

STALL

Causes of Stall:
EXCESSIVE ANGLE OF ATTACK and LOWER AIRSPEED

Stall Recognition:
Aural Warning, Mushy Controls and Buffeting of wings

Stall Recovery:
Reduce Angle of Attack and Increase Airspeed

" THE AIRPLANE WILL STALL AT ANY AIRSPEED AT ANY ATTITUDE"

Spins

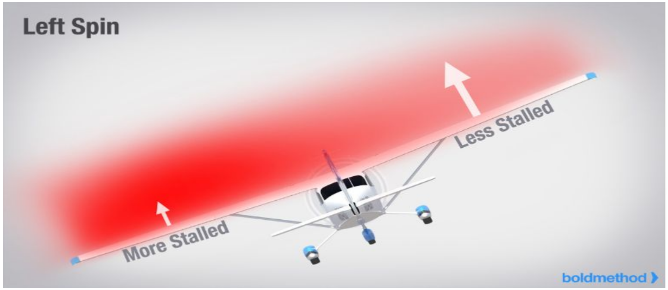
A stabilized spin is not different from a stall in any element other than rotation and the same load factor considerations apply to spin recovery as apply to stall recovery. Since spin recoveries are usually effected with the nose much lower than is common in stall recoveries, higher airspeeds and consequently higher load factors are to be expected. The load factor in a proper spin recovery usually is found to be about 2.5 Gs.

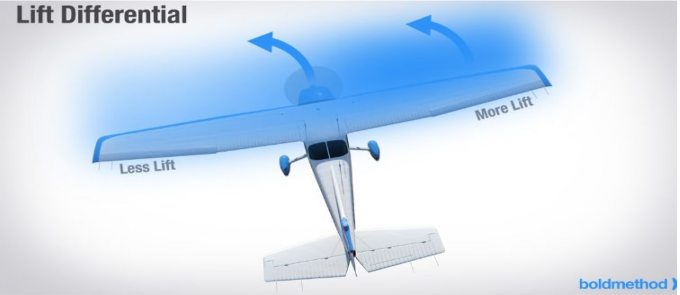
The load factor during a spin varies with the spin characteristics of each aircraft, but is usually found to be slightly above the 1 G of level flight. There are two reasons for this:

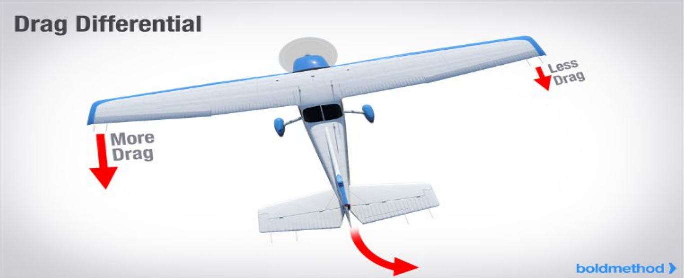
1. Airspeed in a spin is very low, usually within 2 knots of the unaccelerated stalling speeds.
2. An aircraft pivots, rather than turns, while it is in a spin.

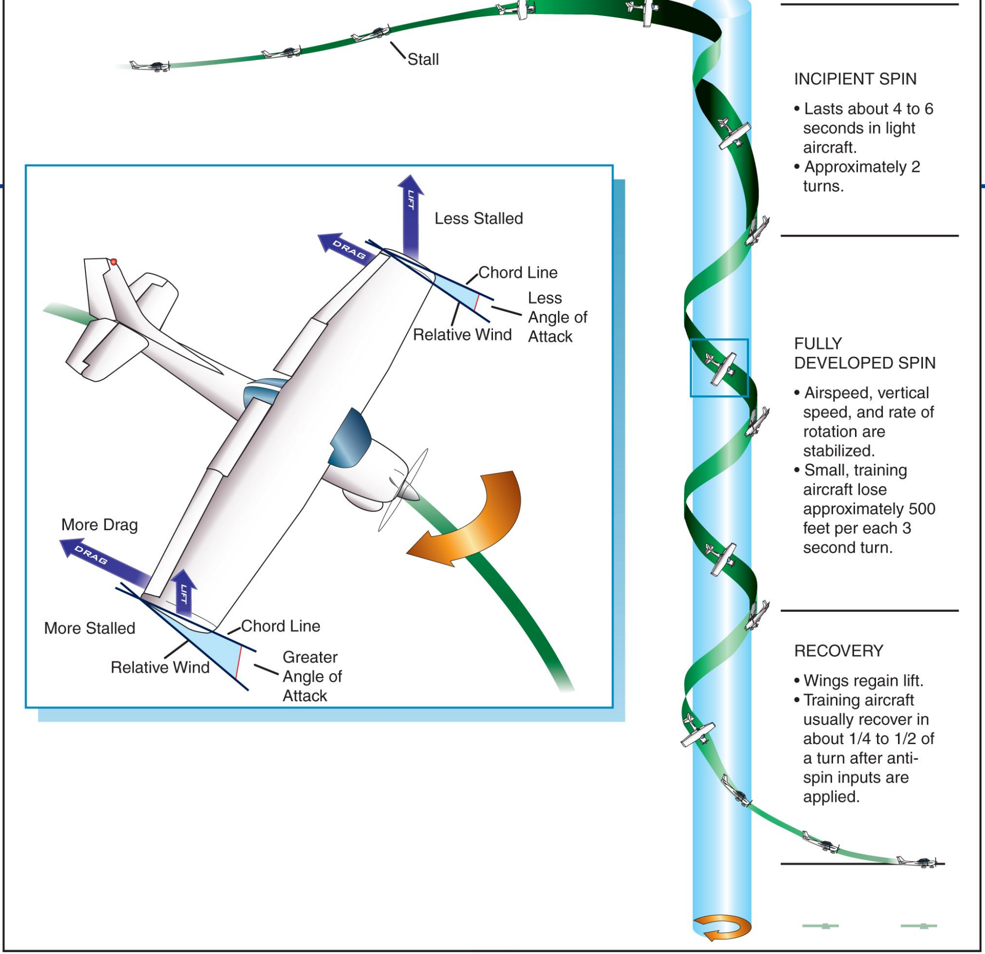
Spins

Since your high wing generates more lift than the low wing, it rolls your aircraft into the spin. And at the same time, your low wing produces more drag, because it's at a higher angle-of-attack. And that drag causes your plane to yaw into the spin. When you combine both forces, you wind up in a fully-developed spin.








Stall
Less Stalled
LIFT
DRAG
Chord Line
Less Angle of Attack
Relative Wind
More Drag
DRAG
LIFT
More Stalled
Chord Line
Relative Wind
Greater Angle of Attack
INCIPIENT SPIN
• Lasts about 4 to 6 seconds in light aircraft.
• Approximately 2 turns.
FULLY DEVELOPED SPIN
• Airspeed, vertical speed, and rate of rotation are stabilized.
• Small, training aircraft lose approximately 500 feet per each 3 second turn.
RECOVERY
• Wings regain lift.
• Training aircraft usually recover in about 1/4 to 1/2 of a turn after anti-spin inputs are applied.

SPIN AVOIDANCE

- Spin Awareness In order to spin any airplane, a stalled condition needs to exist. At the stall, the presence or introduction of a yawing moment can initiate spin entry.

- A spin is an aggravated stall condition that may result after a stall occurs. Mishandling of yaw control during a stall increases the likelihood of a spin entry

SPIN RECOVERY

To recover from intentional or inadvertent spin, use the following procedure:

1. Retard throttle to idle position.
2. Apply full rudder opposite to the direction of rotation.
3. After one-fourth turn, move the control wheel forward of neutral in a brisk motion.
4. As the rotation stops, neutralize the rudder, and make a smooth recovery from the resulting dive.

STRESS LOAD ON GROUND

- Many forces and structural stresses act on an aircraft when it is flying and when it is static. When it is static, the force of gravity produces weight, which is supported by the landing gear. The landing gear absorbs the forces imposed on the aircraft by takeoffs and landings
- Stresses on the wings, fuselage, and landing gear of aircraft are tension, compression, shear, bending, and torsion. These stresses are absorbed by each component of the wing structure and transmitted to the fuselage structure.
- The empennage (tail section) absorbs the same stresses and transmits them to the fuselage. These stresses are known as loads, and the study of loads is called a stress analysis.